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EGE UNIVERSITY

Graduate School of Applied and Natural Science



ALZHEIMER DISEASE DETECTION WITH ARTIFICIAL LEARNING TECHNIQUES

MSc Thesis

Yaren GUNDUZ

Department of Computer Engineering

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Yaren GUNDUZ

Supervisor : Dr. Özgür Gümüş

Department of
Computer Engineering
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Yaren Gündüz tarafından yüksek lisans tezi olarak sunulan “Alzheimer Disease Detection with Artificial Learning Techniques” başlıklı bu çalışma EÜ Lisansüstü Eğitim ve Öğretim Yönetmeliği ile EÜ Fen Bilimleri Enstitüsü Eğitim ve Öğretim Yönergesi’nin ilgili hükümleri uyarınca tarafımızdan değerlendirilerek savunmaya değer bulunmuş ve 7 Nisan 2023 tarihinde yapılan tez savunma sınavında aday oybirliği/oyçokluğu ile başarılı bulunmuştur.

Jüri Üyeleri:

İmza

Jüri Başkanı

: Dr. Özgür Gümüş

.....

Raportör Üye

: Prof. Dr. Aybars Uğur

.....

Üye

: Doç. Dr. Korhan Karabulut

.....

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EÜ Lisansüstü Eğitim ve Öğretim Yönetmeliğinin ilgili hükümleri uyarınca Yüksek Lisans Tezi olarak sunduğum “Alzheimer Disease Detection with Artificial Learning Techniques” başlıklı bu tezin kendi çalışmam olduğunu, sunduğum tüm sonuç, doküman, bilgi ve belgeleri bizzat ve bu tez çalışması kapsamında elde ettiğimi, bu tez çalışmasıyla elde edilmeyen bütün bilgi ve yorumlara atıf yaptığımı ve bunları kaynaklar listesinde usulüne uygun olarak verdiğimi, tez çalışması ve yazımı sırasında patent ve telif haklarını ihlal edici bir davranışımın olmadığını, bu tezin herhangi bir bölümünü bu üniversite veya diğer bir üniversitede başka bir tez çalışması içinde sunmadığımı, bu tezin planlanmasından yazımına kadar bütün safhalarda bilimsel etik kurallarına uygun olarak davrandığımı ve aksinin ortaya çıkması durumunda her türlü yasal sonucu kabul edeceğimi beyan ederim.

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Adı-Soyadı: Yaren GÜNDÜZ



ÖZET
YAPAY ÖĞRENME TEKNİKLERİ İLE ALZHEİMER
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Bu tezde Alzheimer hastalığının erken teşhisi için hasta bireylerle hasta olmayan bireyleri birbirinden ayırabilen bir sistem tasarlayıp hastanelere sunulacak bir hizmet haline dönüştürülmesi hedef alınmıştır.

Bu tez kapsamında, açık kaynaklardan elde edilen hem tabular formatta klinik verileri hem de görüntü formatında MRI verileri üzerinde çift yönlü araştırmalar yapılmıştır. Klinik verileri ile sınıflandırma ve regresyon makine öğrenim modelleri ile çalışılırken, görüntü verileri ile derin öğrenme mimarileri üzerinde çalışmalar yürütülmüştür. İlgili veri kümeleri için algoritmalarından elde edilen sonuçlar farklı metrikler ile karşılaştırılmıştır.

Klinik verilerde çalışmalara başlandığında, öncelikli olarak keşifsel veri analizi yapılmış ve gözlemler içerisinde özelliklere dair boş değerler saptanmıştır. Bu boş verilerin özelliklerle arasındaki korelasyonlar incelendikten sonra korelasyon oranı yüksek olan ilişkiden yola çıkılarak boş değerler impute edilmiştir. Aynı algoritmada bu boş değerler silinerek de çalışma yürütülmüştür, böylelikle imputasyon yapmanın bu verileri ne kadar etkilediği üzerine de çalışmalar sağlanmıştır. Veri ön işleme adımları yapılmıştır ve gözlemler içinde tıbbi özellikler içeren kolonlar arasındaki dağılımlar incelenmiştir. İkili korelasyonlara dair görselleştirmeler yapılmıştır ve dağılımlar incelenmiştir. Çeşitli sınıflandırma algoritmaları ve lojistik regresyon algoritması ile çalışılmıştır. Hiperparametreler

tanımlayarak makine öğrenimi modelleri üzerinde daha iyi sonuçlar elde edilmesi amaçlanmıştır. Yapılan çalışmalardan sonuçlara istinaden klinik verilerdeki en iyi doğruluk oranı AdaBoosting algoritması olarak %86.84 olarak gözlenmiştir.

Görüntü verilerinde çalışırken, veri kümesi eğitim-test-doğrulama olarak üç ayrı gruba ayrılmıştır. Veri arttırımı üzerinde çalışmalar yapılmış ancak veri arttırma işlemi derin öğrenme modellerini test ederken aşırı öğrenme problemine yol açtığı için tercih edilmemiştir. Bu yöntemden uzaklaşıp, eğitim ve test verileri üzerinde bazı ön hiperparametreler tanımlanıp, derin öğrenme modellerinden önce tanımlanarak derin ağda aşırı öğrenme yani ezberleme probleminden uzaklaştırılmıştır. Bu işlem sayesinde modelin ezberleme durumu aşılmış ve daha iyi öğrenme durumu ortaya çıkmıştır. Derin öğrenme modellerinden elde edilen sonuçlar test verileri ile test edilmiştir. Görüntü kümelerinde yapılan sonuçlara istinaden en iyi algoritmanın evrimsel sinir ağı olduğu ve %81.34 olduğu görülmüştür. Yapılan çalışmalar boyunca veri gözlem sayısı ve çeşitliliği azlığından dolayı süreç içerisinde karşılaşılan bazı güçlükler olmuştur. Bu güçlükler doğruluk oranlarını da etkilemiştir.

Anahtar sözcükler: Sınıflandırma Problemleri, Derin Öğrenme, Makine Öğrenmesi, Alzheimer Hastalığı.

ABSTRACT
ALZHEIMER DISEASE DETECTION WITH ARTIFICIAL
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Gündüz, Yaren

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In this thesis, it is aimed to design a system that can distinguish between ill individuals and non-patients for the early diagnosis of Alzheimer's disease and turn it into a service to be offered to hospitals.

Within the scope of this thesis, two-way studies were conducted on both clinical data in tabular format and MRI data in image format obtained from open sources. While working with clinical data with classification and regression machine learning models, studies were carried out on deep learning architectures with image data. The results obtained from the algorithms for the relevant datasets were compared with different metrics.

When studies were started on clinical data, primarily exploratory data analysis was performed and null values were determined within the observations. After examining the correlations between these null data and features, null values were imputed based on the relationship with a high correlation rate. In the same algorithm, the work was carried out by deleting these null values, so that studies were conducted on how much imputation affects these data. Data preprocessing steps were performed and the distributions between the columns containing medicinal properties within the observations were examined. Visualizations of binary correlations were made and distributions were examined. It has been studied with various classification algorithms and logistic regression algorithm. It is aimed

to obtain better results on machine learning models by defining hyperparameters. Based on the results of the studies, the best accuracy rate in clinical data was observed as 86.84% as the AdaBoosting algorithm.

When working on image data, the dataset is divided into three groups as training-test-validation. Studies have been done on data augmentation, but data augmentation has not been preferred because it causes excessive learning problems while testing deep learning models. By moving away from this method, some preliminary hyperparameters are defined on the training and test data, and by defining them before deep learning models, the problem of excessive learning, that is, memorization, is avoided in the deep network. Thanks to this process, the memorization status of the model has been overcome and a better learning situation has emerged. The results obtained from deep learning models were tested with test data. Based on the results made on the image sets, it was seen that the best algorithm was the convolutional neural network and it was 81.34%. During the studies, there were some difficulties encountered in the process due to the low number and diversity of data observations. These difficulties also affected the accuracy rates.

Keywords: Classification Problems, Deep Learning, Machine Learning, Alzheimer's Disease.

FOREWORD

Alzheimer's disease has been an affliction that has eluded humanity's ability to cure for a prolonged period of time. While current diagnostic technologies are sufficient in identifying the disease, no curative treatment method exists to halt its progression. The primary objective of this study is to present a comprehensive investigation where patients diagnosed with Alzheimer's, their caregivers, and medical practitioners can allocate adequate time for diagnosis and establish an effective treatment plan with early diagnosis, thus minimizing the disease's course. By analyzing the data, both clinical and image sets, utilizing artificial intelligence techniques, doctors can diagnose the presence of Alzheimer's disease promptly, thus initiating the treatment process without any delays. The research aims to facilitate early detection of Alzheimer's disease in the realm of health, to bring illumination to the field of medicine and software, and to contribute even marginally to the betterment of human life, thus serving humanity.

07/04/2023

İZMİR

Yaren GÜNDÜZ



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SYMBOLS AND ABBREVIATION LIST

<u>Abbreviations</u>	<u>Explanation</u>
3D	Three Dimensional
3D-CNN	Three-Dimensional Convolutional Neural Networks
AD	Alzheimer Disease
ADNI	Alzheimer's Disease Neuroimaging Initiative
AI	Artificial Intelligence
ANN	Artificial Neural Network
ASF	Atlas Scaling Factor
AUC	Area Under Curve
CNN	Convolutional Neural Network
DCM	Digital Imaging and Communication In Medicine
EDA	Exploratory Data Analysis
EDUC	Years of Education
eITV	Estimated Total Intracranical Volume
FDG-PET	Fluorodeoxyglucose Positron Emission Tomography
FPR	False Positive Rate
GAAIN	The Global Alzheimer's Association Interactive Network
HHMI	Howard Hughes Medical Institute
KNN	K-Nearest Neighbor
kSVM	Kernel Support Vector Machine
LSTM	Long Short-Term Memory
MCI	Mild Cognitive Impairment
ML	Machine Learning
MLP	Multi-layer Perceptron
MRI	Magnetic Resonance Imaging

SYMBOLS AND ABBREVIATION LIST (CONTINUE)

<u>Abbreviations</u>	<u>Explanation</u>
NIFTI	Neuroimaging Informatics Technology Initiative
NRG	Neuroinformatic Research Group
nWBV	Normalize Whole Brain Volume
OASIS	Open Access Series of Imaging Studies
PET	Positrons Emission Tomography
RBF	Radial Basis Function
RF	Random Forest
RGB	Red-Green-Blue Colors
RNN	Recurrent Neural Network
RNN	Recurrent Neural Networks
ROC	Receiver Operating Characteristic Curve
SES	Socioeconomic Status
sMRI	Structural Mild Cognitive Impairment Images
SPECT	Single-photon Emission Computed Tomography
SVD	Singular Value Decomposition
SVM	Support Vector Machine
TANN	Trained Artificial Neural Network
TPR	True Positive Rate
VGG	Visual Geometry Group
VGG Net	Visual Geometry Group Network

1. INTRODUCTION

In human life, individuals share their genes, memories, and thoughts with the next generations, creating a means of self-preservation and communication. As social creatures, humans have communication algorithms to facilitate daily activities, which allows for the transfer of information to distinguish one day from the next. Ancestors encourage the sharing of memories with their grandchildren, passing on culture and life experiences, even after their death. Brain dysfunction can lead to communication breakdown, and while some treatments exist, technology limits their efficacy. Alzheimer's disease, in particular, has a gap in clear treatment, only palliative methods to slow its progression. Early diagnosis is crucial for both patients and their families.

This study aims to provide an early diagnosis of Alzheimer's disease, utilizing both clinical and image data. Clinical data preprocessing, analysis, feature engineering, visualization, and model development were performed to classify patients into different stages of the disease. Many clinical and social features were examined, including education status, socio-economic status, brain atlas factors, and brain volumes. Correlation and distribution analysis revealed null values and, which were imputed according to the most related properties in the data frame. Machine learning libraries, such as NumPy, pandas, scikit-learn, and data visualization libraries, were used to perform exploratory analysis, correlation analysis, and encoding.

Image data preprocessing, analysis, and visualization were also performed utilizing TensorFlow and Keras for machine learning operations. Images were adjusted to a standard dimension, split into train-test-validation, and labeled using one hot label encoding. Different deep learning methods, including recurrent neural network, convolutional neural network, and very deep convolutional network, were used for classification.

The study concluded that the Ada Boosting algorithm provided the best results for the clinical data, while convolutional neural network method was most

effective for image data. Early diagnosis of Alzheimer's disease is crucial for better management and planning of patient care, and the combination of clinical and image data with machine learning techniques can provide a powerful diagnostic tool.

The necessary data for this research were obtained from open-source databases, the methods and procedures were researched and applied during the research process. This detailed information can be found in each section in detail, as indicated in the clauses added below.

Part 1: General information about the thesis was given.

Part 2: Research has been done in the literature about the thesis. In the literature, the researches in this field were examined and studies were made on the methods.

Part 3: Material and Method: Datasets, Applied Operations, Tools, Data Preprocessing and Machine Learning methods. In this section, where the collected data came from, how it was taken, what the data observations included, what the tools were used, how the data preprocessing steps were done, how the models were set up and which methods were processed are mentioned in this section.

Section 4: It is the section where the results are recorded and reported. In this section, the studies and the metrics evaluated together with them were examined and comparisons were made. The connection was established on the algorithm that gave the best accuracy rate and had the best metrics, and the reason was mentioned.

Part 5: The results obtained and their contribution to the literature, what can be done as future work in the literature, going through the results in general.

2. LITERATURE REVIEW

As far as we know, there is no cure for Alzheimer's disease in the scientific community. However, as in every disease, making the diagnosis on time and starting the treatment process early is one of the most important decisions that every doctor will make for the patient. It is known that Alzheimer's disease results in an inevitable death, but timely diagnosis is provided to keep this process longer for both the patient and their relatives.

Many different methods are used to diagnose Alzheimer's disease. These methods are listed as follows;

- Data Obtained Based on Patient History
- Cognitive Tests
- Medical Imaging Methods
- Brain Images

Among these methods, the most accurate diagnosis method is brain images taken by injecting some chemical substances into the body (Nalçakan, 2019).

In the literature, there are many conference articles, publications, and articles that have been studied for the diagnosis of alzheimer's disease.

According to the article of Chu, (2012), the population of the dementia people will increase significantly in the year 2040. The research studies depend on the nutrition of human diets, human brain factor, bioidentical rates, ages, family history, education level, genders and drinking habit. The research says, these factors can affect the ability of alzheimer disease. To relate the phase of the disease, the overall treatment process changes depends on the disease stages. The risk factors and disease rates are calculated according to the features which are studied. What

is important is how early doctors can diagnose this disease and how quickly doctors can start treatment.

MRI (Magnetic Resonance Imaging) is accepted as one of the common techniques used for diagnosing Alzheimer's disease. In studies on MRI data, the shape and volume structures of the obtained brain images were controlled, and image segmentation was applied. Each image fragment formed after the process is subjected to feature extraction with the P-type Fourier equation. As a result of these processes, the accuracy rate achieved in the classification studies is observed as 87.1% (Fuse et al., 2018).

This article explores the use of deep learning-based diagnosis for Alzheimer's Disease, which is a neurodegenerative disorder that has become a significant healthcare concern. Deep learning techniques have been shown to be more accurate than conventional machine learning models for AD (Alzheimer's Disease) diagnosis, making them an appealing alternative for early detection and intervention. This article provides an in-depth review of the current literature on deep learning-based diagnosis of AD, discussing the different deep learning techniques used, such as feed-forward deep neural networks, convolutional and hybrid deep learning models. The study also reviews various biomarkers and datasets used for AD diagnosis, including magnetic resonance images and other images which are provided from technical tools, speech transcripts, genetic measures, and measures. The authors analyze the trends and key findings from the literature, highlighting the challenges that need to be addressed, including data quality, interpretability, transparency, and reproducibility. Despite these challenges, deep learning-based diagnosis shows promise as an effective tool for early detection and intervention in AD (Saleem et al., 2022).

In another article, it provides a comprehensive review of the current state-of-the-art in machine learning techniques for the diagnosis of Alzheimer's Disease. AD is a neurodegenerative disorder that affects millions of people worldwide, making

early detection and intervention critical for managing the disease. Machine learning techniques have shown promise for the early diagnosis of AD by analyzing and identifying patterns in various biomarkers, including brain imaging, genetic data, and clinical measurements. The article reviews various machine learning models, including support vector machines, random forests, neural networks, and deep learning models, and discusses their respective strengths and limitations. The authors highlight the need for robust and diverse datasets for AD diagnosis and discuss the different available datasets. Finally, the article discusses the challenges that need to be addressed to enable the effective application of machine learning techniques in AD diagnosis, such as data quality, interpretability, and transparency. Overall, this review demonstrates the potential of machine learning techniques in AD diagnosis and highlights the need for further research and development to overcome the existing challenges. (Tanveer et al., 2020).

This article reviews the current state-of-the-art in using machine learning techniques for the diagnosis of Alzheimer's Disease (AD) based on neuroimaging data. The authors provide an overview of the different machine learning models that have been used, such as support vector machines, random forests, and artificial neural networks, and discuss the various neuroimaging modalities that have been employed, including structural and functional magnetic resonance imaging (MRI), PET (Positron Emission Tomography), and SPECT (Single-photon Emission Computed Tomography). The article also covers the different biomarkers that have been identified as potential indicators of AD, such as brain volume, cortical thickness, and amyloid-beta deposition. The authors highlight the importance of using machine learning techniques to integrate these biomarkers and improve the accuracy of AD diagnosis. They also discuss the challenges associated with developing and validating machine learning models for clinical use, including the need for large and diverse datasets, standardized protocols for data acquisition and processing, and robust validation methods. Overall, this review provides valuable insights into the potential and challenges of using machine learning techniques for AD diagnosis based on neuroimaging data (Lama et.al, 2017).

In the study of Cui et.al, (2019), Alzheimer's disease is a serious public health concern affecting an estimated 24 million people worldwide. Diagnosis of Alzheimer's disease is a challenging task, as it is difficult to detect early symptoms such as mild cognitive impairment. However, studies have shown that mild cognitive impairment has a high chance of developing into Alzheimer's disease. To aid in early diagnosis, this paper proposes a deep learning approach using an improved Inception neural network to accurately diagnose patients with mild cognitive impairment or Alzheimer's disease based on brain magnetic resonance images. The approach is tested on an international Alzheimer's disease dataset, and the results show that the total accuracy of the proposed approach is 85.7%.

The researched article in literature discusses the challenges in identifying early MCI (Mild Cognitive Impairment) as a precursor to Alzheimer's disease, and how machine learning algorithms have been applied to address this issue. The paper proposes a hybrid Deep Learning Approach that combines multimodal imaging and Convolutional Neural Network with the Long Short-term memory algorithm to detect early AD. The proposed methodology achieved an accuracy of 98.5% in classifying cognitively normal controls from another technology, which suggests that deep neural networks can automatically discover imaging biomarkers indicative of AD and accurately identify the illness. The paper emphasizes the importance of early detection and treatment of AD and highlights the potential of machine learning algorithms to improve diagnosis and patient outcomes (Balaji et al., 2023).

Another paper proposes a deep learning-based approach for diagnosing Alzheimer's disease from brain MRI images. The authors highlight the significance of AD as a global health concern and emphasize the need for accurate diagnosis for effective treatment. They discuss the limitations of traditional methods of AD diagnosis and suggest that deep learning algorithms can improve accuracy by

learning complex representations from large datasets. The proposed method uses a CNN (Convolutional Neural Network) architecture to extract features from MRI images, followed by a MLP (Multi-layer Perceptron) for classification. The authors evaluate the proposed method on a publicly available dataset and compare it with other state-of-the-art methods. The results show that the proposed method outperforms the compared methods, achieving an accuracy more than 90% in classifying AD patients from healthy controls. The authors conclude that their method can provide a reliable and efficient tool for early diagnosis of AD, which can help in the development of effective treatments (Suganthe, et.al, 2020).

Hong et al., (2019) use the LSTM (Long Short-Term Memory) autoencoder to predict the progression of dementia in MCI subjects in order to make an early diagnosis of the disease. These procedures determine the disease's risk level based on data obtained from the ADNI (Alzheimer Disease Neuroimaging Initiative) database, and it is seen that a regression model based on disease progression estimation has been created. An automatic learning model with dynamic parameters is introduced using this approach.

Biomarkers are as important as other markers for the diagnosis of the disease. Studies are carried out on the data obtained from the electroencephalograph signals. These signals can be defined as the main marker for the disease. Signals are first placed in the database and then extracted from the database by independent component analysis. Feature extraction is applied to the data with Wavelet transform. Classification is carried out with SVM (Support Vector Machine). In addition, it is seen that a monitoring mechanism has been developed in which individuals with Alzheimer's disease can travel without their assistants by making an application that includes this method (Thakare et al., 2016).

The SVM method is seen as the most preferred method in research with tabular format data in this field, at the same time, deep learning or neural networks are

generally preferred in research with image data. In another thesis study conducted in this field, data is obtained from the ADNI database. Nalçakan, (2019) defines MRI image data as multidimensional matrices throughout the study process. The model was developed in four basic steps using convolutional neural networks. There are 4 layers in total in the neural network;

- Input Layer
- Convolution Layer
- Sampling Layer
- Fully Context Layer

Deep learning techniques are used for the classification process of carrying and not carrying the disease among MRI images. Depending on the results, the accuracy rate is determined as 90.6%.

It aims to reduce the cost of the alzheimer's identification process by applying machine learning techniques. Being able to correctly diagnose and properly stage Alzheimer's disease can prolong a patient's survival time, as mentioned earlier. In this case, in the MCI stage, to be able to make impressions about the disease, initiate treatment, and collect data is said to be one of the most important steps. At this stage, the biomarkers found were determined, and a study was carried out on the improvement of the process with cost, performance, and time consumption analyzes related to them (Rossini et al., 2020).

Although this disease requires working on clear data, it is possible to obtain constructive results on the diagnosis prediction by using psychological parameters. To achieve this goal Neelaveni et al., (2020), in the study conducted, the prediction of Alzheimer's disease was diagnosed by machine learning techniques based on age at hospital arrival, education level, and other information. psychological

number. SVM is seen as a model that can parse objects using a hyperplane. It is used for both classification and regression. The main advantage of using this algorithm is to be able to distinguish linear or non-linear data. Thus, the diagnosis of the disease is made easier. The research was supported by the R language. The caret and e1071 packages offered by the R language for the algorithm were used. The psychological parameters were processed by both SVM and decision trees algorithm.

It has been a learning model and it can easily solve all kinds of problems. Every little change made to the data can lead to much bigger changes in the future. The packages used here are party, rpart and rpart.plot. When the results are examined, it is seen that the decision support tree algorithm was defeated by SVM 83% to 85%. The developed model can predict the disease and cognitive impairment in the patient and make a distinction.

A study was conducted to provide an automatic diagnosis of Alzheimer's on data from individuals over the age of 18 years, containing 416 images obtained by Dr. Randy Buckner, Ali, (2016). In this study, a similar but different technique to the above studies was developed. TANN (Trained Artificial Neural Network) has been a method to achieve high performance compared to other classification techniques. Compared to other studied classification techniques, the accuracy rate with the trained neural network was 99.2%. According to the opinions of the people working on the research, if Bayesian networks are included in the development, a better working model is produced. Future research studies include the application of Bayesian networks for relevant feedback and more extensive testing with other examples of forensic image studies. It is also envisaged to conduct subjective tests with inputs from forensic medicine experts.

In another research, SVM determines a distributed strategy by applying ANN (Artificial Neural Network), Deep Learning methods. The data on which SVM techniques are most studied are observed as sMRI (Structural MRI -

Structural Mild Cognitive Impairment Images). Feature selection and extraction of the data play an important role in classification. Different data are combined with other data by feature extraction method. Testing the multi-kernel method in SVM has increased the accuracy values considerably (Tanveer et al., 2020).

Although studies for early diagnosis of alzheimer's disease are at the forefront, controlling the behavior of the patient during the treatment process and knowing which treatment may be appropriate according to the remaining age life is problematic in some painful situations, but also requires speed. For this reason, the calculation of the remaining life of the patient is also important for better adaptation of the patient to the treatment process rather than the psychological dimension of the condition. However, with such analyzes, the right treatment can be applied to the patient and the life span can be extended according to the course of the disease. Deep learning methods were investigated in research conducted on this subject in the literature.

The remaining age calculation of the patient has been provided. To build a new convolutional model to predict survival in a longitudinal context, using alzheimer's disease as a case study for experimental validation. The main contributions are threefold:

- A general framework is developed for the longitudinal survival prediction task, which provides a basis for comparing performance across models.
- To account for dynamically updated survival estimates and potentially informative data sparsity models for the proposed architecture.
- A proposed method for visualizing influencing variables to better understand the model's predictions with medical applications.
- Modern results are discussed when compared with a large benchmark group.

This disease is a neurological disease that harms human brains and brains' cells and destroys patients' memories. As a result of the regular destruction of brain cells, the individual becomes unable to meet even his daily basic needs alone. Early diagnosis can always detect the signs in advance and slow down the course of the disease a little bit. In a study similar to the method we proposed Soni and Vahora, (2022), researchers studied support vector machines, decision trees, KNN (K-Nearest Neighbor) and Random tree, CNN, RNN (Recurrent Neural Network) , AlexNet, and ResNet, VGG Net (Visual Geometry Group Network), Inception Net learning models, and collected data. They provided a comparative conclusion regarding the results. Deep learning techniques have helped produce better results.

The main purpose of many diagnostic studies in the field of health, such as disease diagnosis, early diagnosis, and tumor diagnosis, is to improve the diagnosis process in the health sector. For this, early diagnosis is of great importance. In this study Vidushi et al., (2022), research that is very similar to the method we propose emerges. In this study, learning models such as Random Forest which is a combination of Logistic Regression, Binary Classifier, Decision Tree, AdaBoost and Support Vector Machine were used. The results were observed on three basic parameters. These parameters outperform benchmark values on AUC (Area Under Curve), Recall and Precision results from Decision Trees and other learning models with RF 0.8421. The method proposed in this study, on the other hand, produced better results with a ratio of RF (Random Forest) 0.87, which works on the data we have.

The possibility of developing Alzheimer's disease in people with mild cognitive impairment obtained from MRI images has been studied. Common symptoms have been observed thanks to brain changes in patients and patients with a prodrome. Multivariate analysis applications were made. The data sets used in this study Gabriela et al., (2013) were obtained from both volumetric and cortical thickness measurements. An intensity index was used for hidden structures using orthogonal projection. The closer this severity index is to 1, the more patients are diagnosed, and the closer it is to 0, the more mild cognitive impairment is diagnosed. The results obtained from the data in this study are 112 patients with 64.7% similarity

to Alzheimer's patients and 61 patients with 35.3% similarity to mild cognitive impairment. 173 of these patients turned into Alzheimer's patients. As a result of this study, it has been possible and partially succeeded to identify and diagnose potential Alzheimer's patients.

Brain regions in individuals with Alzheimer's disease cause disorders in managing and practicing functional behaviors compared to normal individuals in the study of Ji et al, (2019) researched. In addition to these determinations, low-frequency (<0.08 Hz) changes are seen in the signals depending on the level of blood-oxygen ratio. The researchers divided the entire brain into 116 regions and identified abnormal connections by comparing the correlation coefficients of each pair. see watching. It stems from the reduction of correlations in the fields studied in the proposed and referenced methods.

In another study, a resource was created that combined MRI and PET scans of 150 water tests from 30 people with mild cognitive impairment, which is one of the stages of the disease, and allowed them to be analyzed into a single denominator. Evaluated by cross-validation technique with accuracy up to 100, external cross-validation technique for classification rate and packing cross-validation for predictions. The area under the ROC (Receiver Operating Characteristic Curve) curve was found to be 97.8%. Spatial maps of classification contributions are temporal, prefrontal, orbital, and apical regions. This study of Yong et al., (2008) is one of the best in the literature for detecting abnormal brain behaviors in mild cognitive impairment.

Brain images are a guiding resource for understanding diseases in the neural dimension. It can be said as a valuable resource from which we can easily understand and interpret neurological disorders. In this study of Hinrichs et al., (2009) machine learning techniques were used to diagnose new subjects from the ADNI database in detail with MRI and FDG-PET (Fluorodeoxyglucose Positron Emission Tomography) images. Support Vector Machines, one of the machine learning algorithms, proposes a new method using Linear Program Boosting. While

training the algorithm to classify, it aimed to provide a structure that would decide that it should prefer regions over spatial parts from the sample images in its hands.

Based on gray matter volumes, 69 Alzheimer's patients and 60 normal control topics and aims were included in the study, and a sparse logistic regression classifier was applied due to the small amount of data. Two different methods were applied with this algorithm. Comparisons were made with punished logistic regression with maximum uncertainty linear discrimination analysis. It was observed that smooth disease diagnosis achieved better results with a spatially arranged formulation (Anil et al., 2011).

There appears to be a bottleneck in diagnostic performance for the expression of neuroimaging biomarkers along with the diagnosis of alzheimer's disease by computer-aided machine learning techniques in the study of Siqi et al., (2015). A new diagnostic study with a deep learning architecture designed. It extracts key patterns from some data and reveals additional information. Improvements were observed in binary and multiclassifier performance in disease diagnosis.

In another imaging study Xiaofeng et al., (2016), it was not possible to determine the correct diagnosis rate due to the conflict between high feature size and low sample size. Linear discriminant analysis and locally conservative subspace learning methods have been used in different fields to select class discrimination and noise tolerance features. Experiments on the neuroimaging data set support the proposed method of the study.

A new method Latha et al., (2019) is proposed for the selection of diseased tissues on images for disease diagnosis. First, sagittal, coronal, and axial slices of images are converted into small segments and similar regions are marked. Then feature inference is done with the feature selection algorithm. Random forests, support vector machines, and K-nearest neighbor techniques are used. The

proposed features were graded on baseline MRI images of 812 subjects and resulted in specificity of 89.58% and 85.82%.

This study Hongfei et al., (2019) has been proposed to detect features of potential disease and to capture abnormal conditions of the brain with deep learning architectures in the dataset of ADNI MRI images and records of individuals with the disease and mild cognitive impairment. First of all, to be able to diagnose at the highest-level during training, it is aimed that each layer communicates with the next layers and is directly connected. With the probability-based fusion method, they have managed to gather different architectures under a single architecture.

In a study of Raza et al., (2023), based on the magnetic resonance images of Alzheimer's disease, researchers created a deep learning model on existing machine learning models, completed their training with cumulative incremental epochs such as 10, 25 and 50 in this deep network, and tested the results. According to the results, the model has an accuracy of approximately 97.84%.

Reddy et al. (2023) utilize convolutional neural networks and an information-free network called the nonnegative matrix factorization tensor decomposition network to classify Alzheimer's disease and predict clinical scores based on clinical images. Their approach involves using tensor decomposition and multilevel channel banks for test learning, which allows for more scalability than principal component analysis. The study ultimately aims to improve the accuracy of AD diagnosis and clinical score prediction through advanced deep learning techniques and imaging technology.

Resting-state functional connectivity has been widely used in predicting alzheimer's disease, but conventional correlation calculation does not consider different frequency band features, leaving out high-order neurodynamics. To address these challenges, the authors proposed an automated alzheimer disease diagnosis system that creates a high-order neuro-dynamic functional network using

several levels of time-series data. The proposed model uses multiple customized deep learning models, including SVM, K-nearest neighbors, Alexnet, and Inception blocks, achieving an accuracy of 96.61% and AUC of 0.9663 in differentiating disease subjects from normal controls without external feature selection. The results show the potential of the proposed method for the clinical value of disease prediction (Sethuraman et al., 2023).

The use of Machine Learning techniques in computer-aided diagnosis has proven to be an effective tool for various health applications due to its ability to learn accurately from complex data. deep Learning models have been widely used for the 4-way classification of alzheimer disease in recent years and have achieved good results. However, traditional machine learning classifiers such as boosting algorithms and support vector machine, perform either the same or better than deep learning models and require less data for training. This property is particularly useful in medical applications where large labelled datasets are often unavailable. In this study, the authors propose an ensemble classifier model based on ML models and demonstrate its effectiveness in the 4-way classification of disease using the open access dataset. The proposed ensemble classifier achieved an accuracy of 94.92%, which represents a 5% increase in accuracy compared to the individual classifier approach (Shaffi et al., 2022).

In this thesis, a system that can diagnose Alzheimer's disease from the hospital records of Alzheimer's patients and that can help the doctor in the diagnosis process by making necessary analyzes from the patients' clinical and image information has been studied. For the diagnosis of the disease, basically two main studies have been carried out. While the first arm of these studies was performed with clinical data, the second arm was performed with image data. The clinical data were in tabular format with characteristics of patients that embody basic social and medical characteristics, while the image sets were MRI images of the patients. Although the methods applied for these two different studies are quite independent from each other, the general flow is similar. First, if we need to talk about the studies on the clinical side, the study process and what has been done can be summarized as follows.

Clinical data were collected as tabular data from the OASIS (Open Access Series of Imaging Studies) database containing patient and non-patient records. After exploratory analyzes were performed on these data, data preprocessing and correlation analyzes were performed. Data visualizations were applied to see the emerging relationships more clearly, and then the results were observed, and reports were created by using classification machine learning algorithms such as Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, Ada Boosting to classify the target state with hyperparameter definitions.

Image data contains data in classified image format, which is taken from the ADNI database and found collectively according to the degree of disease. After exploratory analyzes were performed on these data, the dataset was divided into train-test-validation, the data preprocessing step was provided, which picture was in which disease stage was diagnosed with one hot label encoding, and some parametric criteria were determined to minimize and prevent overfitting processes. Finally, artificial neural networks were defined, models and layers were created, the data were trained with these models, and the results were evaluated with the help of the data in the test set. Thus, how well the model learned from the training process was tested and accuracy rates were calculated.

As a result of the study, it was observed that the boosting algorithm gave better results in clinical data, while convolutional neural network was found to be more successful in studies on image sets. Since the boosting algorithms strengthen the weak samples, the application of this algorithm in this dataset has resulted in much better results than other classification algorithms and regression algorithms. As the classification algorithm with the highest accuracy, AdaBoosting provided a percentage with 86.84%, while the convolutional neural network in image sets surpassed other deep network models with the best accuracy performance with a percentage of 81.34%.

3. MATERIALS AND METHODS

The prevalence of Alzheimer's disease has led to a significant amount of data being collected from various sources, with a lack of specific treatment options available. To address this issue, the GAAIN, (2020) GAAIN (Global Alzheimer's Association Interactive Network) was established as an online data platform, which provides access to a wide range of Alzheimer's disease data from different organizations. The platform has data from 528,013 patients from 59 data partners around the world, comprising clinical data, blood values, cognitive test values, physical organ measurements, MRI and PET data, as well as genetic information. The diverse data sets in the platform provide valuable information that can be utilized for specific studies based on their structure through machine learning and deep learning algorithms.

Contributions from OASIS and ADNI databases have been significant in populating the GAAIN platform with data. The imaging and clinical sets were obtained from the Disease Neuroimaging Initiative database.

Access to the data provided by OASIS and ADNI requires submission of a confidentiality form to OASIS and a detailed research idea along with a signed use and confidentiality form to ADNI. After acceptance of the application and signing of the privacy policies, researchers can directly download certain datasets for OASIS or direct them to an auxiliary internal site called ADNI IDA, containing various types of data. Researchers can then select appropriate datasets for the features and criteria to be used in their study, which are available in NIFTI (Neuroimaging Informatics Technology Initiative), DCM (Digital Imaging and Communication In Medicine), and tabular formats.

In this study, the clinical data of Alzheimer's patients were accessed from the OASIS database, and image MRI data were obtained from the ADNI database.

The study focused on data preprocessing, data visualization, and machine learning techniques, employing NumPy, Pandas, Matplotlib, Seaborn, and Scikit-Learn libraries. The classification algorithms, including Logistic Regression, Support Vector Machines, Decision Tree, Random Forest, and Ada Boosting, were utilized for training and testing the clinical data, rather than clustering algorithms. For the image data, the TensorFlow library and Keras framework were used, and deep learning networks were designed to match the relevant pictures in the image sets with the disease level. The data were subjected to the training, test, and validation phase using three artificial neural networks, namely convolutional neural network, recurrent neural network, and very convolutional neural network. In addition to the one-hot-label encoding process, the networks were designed to match the relevant pictures in the image sets with the disease level, providing an effective method for disease diagnosis.

3.1. Data Set Organization

The Open Access Imaging Studies Series is a project aimed at making brain MRI datasets freely available to the scientific community. Scientists hope to facilitate future discoveries in basic and clinical neuroscience by compiling and freely disseminating MRI datasets. The NRG (The Neuroinformatic Research Group) and Biomedical Informatics Research at the University of Washington School of Medicine make available OASIS, University of Washington Alzheimer's Disease Research Center, HHMI (Howard Hughes Medical Institute) at Harvard University, and Dr. Randy Buckner. It appears as a dataset resulting from the compilation of OASIS sets 1, 2, and 3. This dataset contains;

Cross-sectional MRI Data: Data from cross-sectional MRI in young, middle-aged, non-demented, and elderly dementia patients: This dataset includes 416 subjects ranging in age from 18 to 96 years. There are scan sessions included. The subjects were all right-handed and male and female. 100 of the subjects over the age of 60 were clinically diagnosed with very mild to moderate alzheimer's

disease. A safety dataset of 20 dementia-free subjects screened at a follow-up visit within 90 days of their initial session was also included.

Longitudinal MRI Data: Longitudinal MRI data in non-demented and dementia-affected older adults: This dataset includes 150 subjects ranging in age from 60 to 96 years. Each subject was screened twice, at least a year apart, for a total of 373 imaging sessions. Three or four individual T1-weighted MRI scans obtained in single scan sessions are included for each subject. The subjects were all right-handed and male and female. Throughout the study, 72 of the subjects were classified as dementia-free. Sixty-four of the subjects were diagnosed with dementia at the time of their first visit and remained so at subsequent screenings, including 51 with mild to moderate Alzheimer's disease. The other 14 subjects were classified as dementia-free at the time of the study.

In the image sets obtained through ADNI, it contains 6400 MRI images of patients over 4 main target classes. These images were taken in raw format from ADNI without preprocessing and used in this thesis. These images were divided into two classes as train and test and the study was carried out using deep learning techniques. These target classes can be briefly stated as follows.

- Mild demented
- Moderate demented
- Non-demented
- Very Mild Demented

Images are available in the datasets both axially, sagittally and coronally in the study.

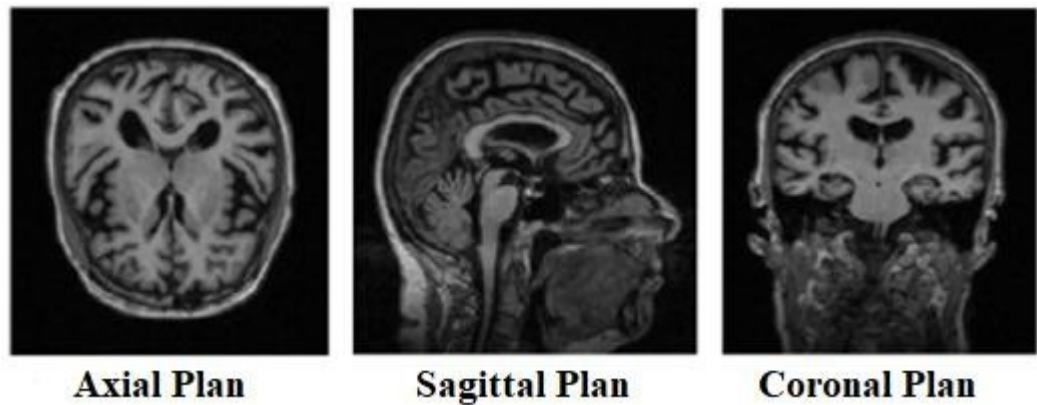


Figure 3.1 Alzheimer Disease Patient MRI Plans (Nalçakan, 2019, p. 21)

Images obtained from MRI devices are in NiFTI or DCM file type. While processing these file types, the train set was trained by converting PNG files to 3D (Three Dimensional) images and applying deep learning techniques, and it was tested how much the model learned with the test set. Feature extractions from these image sets were processed with the help of CNN. After the relevant results, it was tried and seen that the result reached higher accuracy values with a slight difference after trying two or three-layer CNN architectures tried during the thesis period.

The image sets were retrieved sporadically from the database, and there were missing information in the records in the clinical data. Before working on the models during the thesis study process, it is necessary to make sure that the data preprocessing is done in the most correct way and that the data is in the best and cleanest format. Several of the observations in the clinical data had missing data. There were two basic processes to follow here. The first was to remove these missing data from the data frame, and the second was to perform correlation analysis between features and fill in empty data with median, mode, mean, and a feature with the highest correlation. When working on Logistic Regression, two different approaches were created for these two operations. For other classification algorithms, the study was carried out on the imputed data frame. Since there are

records of basically four different disease levels in the image sets from the database, labels were defined according to these disease degrees to the images, thus completing the data preparation part. Data augmentation processes were also tried on image sets, but since the number of data was large, overfitting rates increased a lot while defining these data to the models, so it was determined that the data augmentation process was not suitable for image sets and this process was removed from the study.

3.2. Tools

The use of open-source software libraries and tools in the scientific literature has recently increased due to the popularity of artificial intelligence applications and deep learning studies in the health field. Global giant companies presented their source codes to researchers and developers on version controlling sites. In this way, artificial intelligence studies are increasing day by day.

Although shared plugins and libraries have used for different software languages, Python was chosen as the reason for preference in this thesis, since Python is the most widely used programming language for data science. Since Anaconda is a faster platform that contains many packages, Python codes were written using Jupyter Lab and Jupyter Notebook through the Anaconda distributor. This distributed platform is helpful as Rolon-Mérette et al. (2016) said in their study for the programmers who can write in Python and work for data science also. There is also another environment that is called as Google Colab which provides extra computing engine and is helpful to run the study kernel online with distributed engines too. This system benefits are so much in data science field as Carneiro et al., (2018) mentioned. In addition, the same studies were carried out on the Google Colab Notebook during the working process. The main reason for using Google Colab was that the data was too much and when running with a single computing engine, the working kernel was overloaded and the time to get results was interrupted. Google Colab has the right to work free of charge for a certain period of time. Within this right, the right to compile studies is granted by increasing the

number of computing engines. In the works, the computing engine support via Google Colab was taken and the work continued when the Jupyter Notebook and Jupyter Lab were blocked according to the computing and the huge amount of data in datasets.

First of all, in order to be able to work on Python with these data for the purpose of data science, it is necessary to install the open-source libraries Python offers to data scientists and software developers into the code kernel and computer. NumPy, Pandas for data preprocessing steps; Matplotlib, Seaborn for data visualization steps; For machine learning processes, Scikit-Learn and TensorFlow libraries were installed on the computer and the working process was started. Libraries used in Python codes are as follows for each study in this research.

1. For clinical data in tabular format;
 - For data preprocessing --> NumPy and Pandas
 - For data visualization --> Matplotlib, Seaborn
 - For data training, modeling, and testing --> Scikit-learn, SciPy
2. For clinical patient data in image format;
 - For data preprocessing and feature engineering --> OpenCV, TensorFlow
 - For data visualization --> Matplotlib, Seaborn
 - For data training, modeling, and testing --> TensorFlow and Keras Framework under TensorFlow were used.

3.3. Data Processing

In data science, creating the most appropriate data structure before modeling is crucial as it affects the training and testing rate of the model. The data preprocessing step is the foundation of data science as it significantly impacts a data scientist's output. If the data preprocessing steps are not done well enough, even

good and clean data can produce wrong results. Therefore, data preprocessing is one of the most time-consuming and valuable processes in the data science process.

When starting a project, a data scientist first has data in hand. Exploratory analyzes should be performed to understand the meaning of the data and the story underlying the dataset. The first step is to decide how the data can be used, what it tells, which features are critical, and which observations or features are not needed. Data preprocessing involves reducing the noise in the data, removing or imputing unnecessary values in the observations, extracting new features from the data, establishing relationships between features, and creating a result-oriented data science story.

Data preprocessing is a laborious process that adds value to reach the result. It enables obtaining valuable information from scattered, complex, incomplete, and badly formatted data. A smooth and clean formatted dataset is created through data preprocessing, facilitating a transition to the data visualization and machine learning phase.

Moreover, data preprocessing is a fundamental process that should be performed with great care to produce meaningful and valuable results in data science. It is the first and essential step to extract insights and patterns from raw data. Data scientists should pay close attention to data preprocessing to ensure the creation of a well-formatted dataset that can be used in data modeling, data visualization, and machine learning applications.

The main features for clinical type data are years of education, socioeconomic status, the exam that occurs when suspicious about the disease for the patients, Alzheimer disease detection, estimated total volume of brain structure,

whole brain volume, atlas scales of the head skeleton. *Figure 3.2 also* summarizes the main features in the data frame in tabular format clinical data obtained through OASIS.

Exploratory analysis was performed on the data. After exploratory analysis, empty cells for socioeconomic status were detected between observations. Organizing data properties are the first task to do about this data frame. Eight rows are identified with missing values in the SES (Socioeconomic Status) column. At the same time, correlation analysis between features was performed. It was observed that the feature with the closest relationship with socioeconomic status was the number of years of education. This issue is also clarified with two approaches. The first of the two basic steps is to remove rows of blank cells from the dataset. The other is to replace the missing values with the corresponding values, also known as 'Imputation'. Since in tabular format data, there are only 150 observations, imputation also helped the performance of our model. The rows of these null values from the separate branch were deleted over the data frame and this was saved as a separate data frame. The main purpose of creating two different data frames was to determine how much the imputation process would affect the accuracy rates. Thus, blank values were filled with the median according to the breakdown of this feature on socioeconomic status. This imputation with the median was recorded also as a data frame.

COL	FULL-FORMS
EDUC	Years of education
SES	Socioeconomic Status
MMSE	Mini Mental State Examination
CDR	Clinical Dementia Rating
eTIV	Estimated Total Intracranial Volume
nWBV	Normalize Whole Brain Volume
ASF	Atlas Scaling Factor

Figure 3.2 Main Features in Clinical Dataset

The SES column is also imputed depending on the average value of the education level of each patient. The average graph for SES and EDUC (Years of Education) properties according to the observations. To the story of data science, socioeconomic status is directly related to education level. Before starting the imputation process, a line chart was created to see the relationship between these two features even more closely. It could be seen more clearly in *Figure 3.3*.

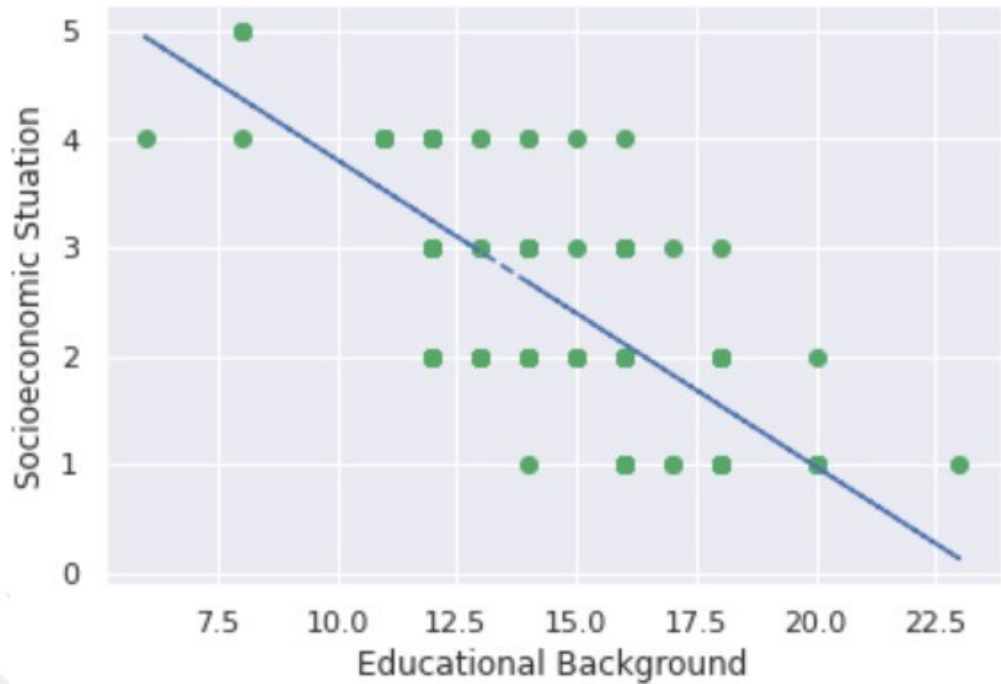


Figure 3.3 The Relation Between Education Level and Social Economic Status

The preprocessing steps on the image data obtained from ADNI database. The ADNI site allows us to download preprocessed MRI images. While working on the image sets, it was worked on separating the images according to the degree of disease. This was done by defining labels by doing one-hot label encoding. Afterwards, the data were divided into 20 cross fold validation rate as train-test-validation sets. Time was spent on the data augmentation process, but after this process, the overfitting problem was encountered in the models, and it was decided that it was not suitable for image sets from the study and was removed.

Data preprocessing steps made with the help of TensorFlow library and Keras framework using Python language;

- Extra-brain parts were examined and clipped with the help of File Viewer for image observations.
- Holistic data is divided into the train, testing, and validation with 20 cross-fold validation rate.
- One hot label encoding was performed to identify which MRI image belongs to which stage of the disease.
- One hot label encoding is implemented for labeling the image data according to their disease stages.
- Prefetch operations were performed on train and validation sets before the CNN definition.

3.4. Implemented Methods

In clinical data, there is a feature called Clinical Dementia Rating in the data frame. This property is the target column in the data frame. The information contained in this behavior includes information that makes the decision whether the patient is ill or not. In this way, study were carried out with classification algorithms on clinical data. There are five basic machine learning algorithms used under this study for the classification process. These algorithms serve as classifier algorithms for the early diagnosis and detection of alzheimer's disease. These machine learning algorithms which are used in this research;

- Logistic Regression (with Imputation and with drop nan values)
- Support Vector Machine
- Decision Tree
- Random Forest
- AdaBoost

Hyperparameters are crucial in machine learning, as they can significantly affect the performance and accuracy of the model. Hyperparameters are set prior to

the training process and control various aspects of the learning algorithm, such as the learning rate, regularization strength, number of hidden layers, activation functions, and more. By tuning the hyperparameters, the goal is to find the optimal configuration that maximizes the performance of the model on a given dataset.

However, hyperparameter tuning can be a complex and time-consuming process, as it often requires a trial-and-error approach to find the optimal combination of hyperparameters. This is commonly done by performing a grid search or random search over a predefined range of values for each hyperparameter.

It is worth noting that hyperparameter tuning is a crucial step in the machine learning process, as poorly tuned hyperparameters can lead to overfitting, underfitting, or low performance on unseen data. Thus, careful consideration and experimentation with hyperparameters are necessary to achieve the best results in machine learning. The hyperparameters and values used in clinical data can be listed as follows.

- Logistic Regression:
 - C: 10
- Support Vector Machine:
 - Kernel Type: RBF
 - C: 100
 - Gamma: 0.1
- Decision Tree:
 - Max Depth: 1
- Random Forest:
 - Max Depth: 7
 - Numbers of Features: 5
 - Number of Trees: 2
- AdaBoost:
 - Estimator Number: 2

The hyperparameters and their values used in the image sets can be listed as follows.

- RNN:
 - Optimizer: Adam
 - Epochs: 50
 - Hidden Layer: 3
- CNN:
 - Optimizer: Adam
 - Epochs: 50
 - Hidden Layer: 3
- VGG (Visual Geometry Group):
 - Optimizer: Adam
 - Epochs: 50
 - Hidden Layer: 4

The C parameter serves as a penalty score, allowing for adaptation of the machine learning algorithm by penalizing data as necessary to ensure accuracy. In the case of non-linearly separable classes, a kernel trick is applied, such as the G-RBF Kernel (Gaussian RBF Kernel) found to be the best parameter in this thesis. The gamma parameter is utilized to control the width of the scatter, with a lower value indicating a wider distribution. The gamma parameter should be decreased if the model is overfitting and increased if it is underfitting, as observed in the C hyperparameter.

The Max Depth function returns the maximum depth of a tree-based algorithm, which is the shortest path between the root and the leaf node. In the forest

algorithm, Max Features represents the maximum number of features that the tree-based ML (Machine Learning) algorithm should classify. The inclusion of multiple trees, known as Tree numbers, within the forest ML algorithm leads to improved decision making. The estimator number is applied in boosting algorithms to enhance weak classes and promote more powerful outcomes.

The Adam optimizer is utilized to optimize the ML model in neural networks. The number of epochs is a hyperparameter that governs how many times the training dataset is traversed. Hidden layers, which exist between the input layer and output layer, influence the accuracy of neuron weights to achieve the desired level of accuracy in the neural network model.

In addition to the parameters and their values, it is crucial for a data scientist to select the most appropriate algorithm for their data to obtain the best possible results. To this end, several classification algorithms were utilized and compared in this study, and the results are detailed in the subsequent sections, specifically as applied in clinical and image data sets.

Logistic Regression:

Logistic regression, despite its name, is a classification algorithm that aims to distinguish between multiple samples based on their ancestral class. The biggest difference with linear regression is how it applies the line separating the two classes. That's really how it shapes the pattern. Linear regression uses the "least squares method" to plot the optimal line, while logistic regression uses the "maximum probability of occurrence". Logistic regression is a data analysis technique that uses mathematics to find a relationship between two data elements. Logistic regression then uses this relationship to estimate the value of one of these factors as a function of the other. Predictions often have a limited number of outcomes such as yes or no.

As Sperandei, (2014) says, logistic regression is an important technique in the field of artificial intelligence and machine learning. Machine learning models are software programs that you can train to perform complex data manipulation tasks without human intervention. Machine learning models built using logistic regression help organizations derive actionable insights from business data. They can use this information for predictive analysis to reduce operational costs, increase efficiency and scale faster.

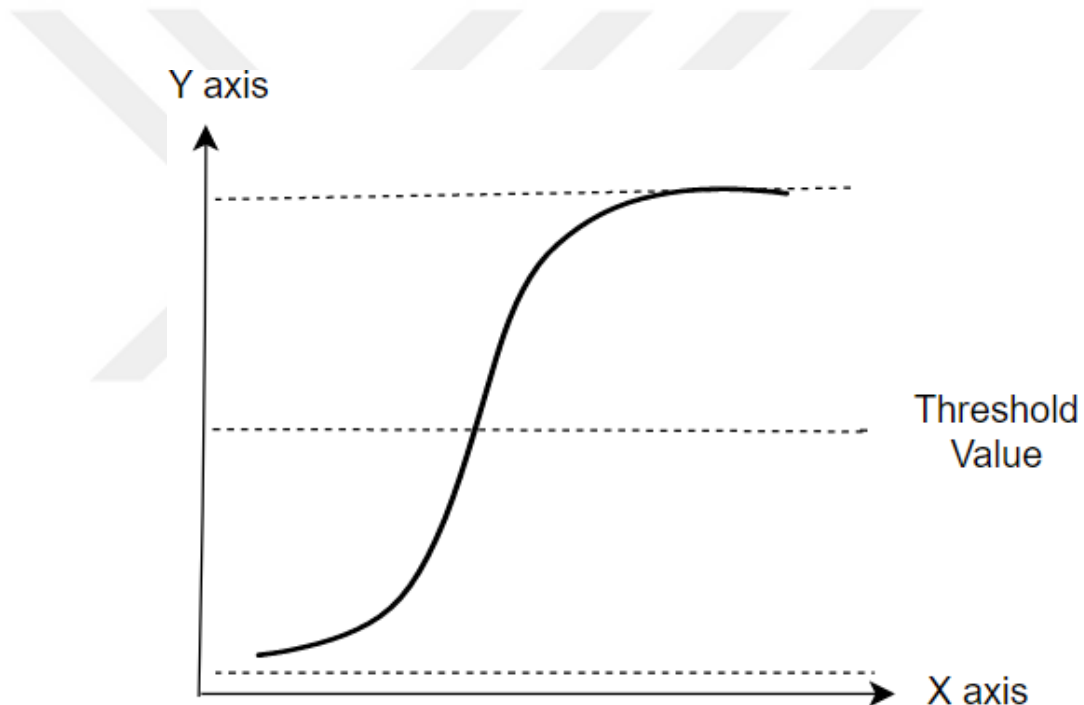


Figure 3.4 The Logistic Regression Curve

Logistics regression uses the sigmoid method for classification. The sigmoid method to compress the obtained data into the range of 0 and 1. The sigmoid function is a mathematical method which has a characteristic “S-Shaped” curve.

$$S(x) = \frac{1}{1 + e^{-x}} = \frac{e^x}{e^x + 1} = 1 - S(-x).$$

Figure 3.5 Sigmoid Function

The sigmoid function is a limited, differentiable real function that is defined for all input real values and whose derivative is not negative at every point and exactly one inflection point. The sigmoid function is a monotonous and bell-shaped function (with a local maximum and no local minimum) derived from its own mathematical expression. Thanks to this function, classification can be made by this local minimum and no-local minimum points. Each method has advantages as well as disadvantages (Stoltzfus, 2011).

Advantages;

- Implementation and interpretation are easy by using this machine learning algorithm.
- It performs quite well if the dataset is linearly separable.
- It is less prone to overfitting but can be overfitted on large datasets.

Disadvantages:

- If count of observation is not big amount, the regression algorithm that is called as logistic regression should not be in the case, otherwise overfitting may occur.
- For logistic regression to discriminate, the data set must be linearly separable.

In literature, according to the mathematics and statistical explorations, Currently, data scientists like Lemeshow et al., (2013) uses three well known regression model kinds when using logistic regression. These are briefly;

- ***Binary Logistic Regression***

- The response, or dependent variable, in this strategy is binary. Only two responses are dependent on a single outcome. For instance, "0" or "1," "male" or "female," "yes" or "no," or "predicting" whether a tumor is malignant, whether a disease is present, or whether a condition has been created. This is the method that is used the most frequently in logistic regression and, more generally, one of the most frequently used classifiers for binary classification.

- ***Multinomial Logistic Regression***

- The dependent variable in this type of logistic regression model has three or more possible outcomes; However, there is no established order for these values. To better market movies, movie studios, for instance, want to know what kind of movie a viewer is most likely to see. The studio can use a multinomial logistic regression model to figure out how much a person's age, gender, and dating status may affect the kind of movie they like. The studio can then target a specific group of potential moviegoers in an advertising campaign.

- ***Ordinal Logistic Regression***

- ○ When the target variable has three or more possible outcomes, this kind of logistic regression model is used. However, in this case, the values have a specific order.
- ○ There is some significance. Like a soldier's grades, professors, associate professors, students in a school, or exam results (A-F) for students, like survey rating scales (like 1-10).

This analytical method could be utilized in medicine to predict and categorize the likelihood of disease in each population.

Logistic regression method is among the methods applied in the research conducted with clinical data. Logistic regression is a very important tool in data science. This algorithm is a classification and regression algorithm that allows estimation of the probability of an event occurring based on the values of the independent variables or classification into one of two categories. It performs its operations over the mathematical function that tries to mark the probability predicted by the target feature in a region between 0 and 1. Although the reasons for its use in this study are generally explained in the sentences above, the aim is to look at the clinical characteristics of the patients and to make predictions to the doctors whether this individual has Alzheimer's disease.

In this thesis, logistic regression is used to classify the Alzheimer's Disease stages from a given population which are gathered in OASIS database as a clinical data. Before working on logistic regression, data were retrieved from the OASIS database as observations in tabular format. There was a column called clinical dementia rating, which included the disease stages of the disease. This column was the target feature column in the dataset that will continue throughout the research. This column is basically a column containing the values 0 and 0.5 and containing numeric type data encoded as ill or not. 0 and 0.5 values in this column were classified by label encoding method as demented and nondemented by denoting 0 and 1. Thus, the numerical information in the target column was made more meaningful. The ordinal logistic regression is mainly used to understand the disease stages' importance and order in the four stages of the disease. There are classes in the target feature of the stage of the disease. The information of this feature has been examined. Primarily, these operations were performed after the exploratory analysis and in the preprocessing stages of the data. Afterwards, before moving on to each machine learning algorithm, hyperparameter definitions were made and the training process was started on the model. The logistic regression process was run by filling in some empty cells by looking at other columns because of EDA (Exploratory Data Analysis), or by removing empty features. Logistic regression

results obtained as a result of two different methods are confined in a table. Although the results obtained because of the two were close to each other, the accuracy rate obtained as a result of the removal of empty cells was seen as 78,94% and without dropping removal cells accuracy result is recorded as 76.31%.

Support Vector Machine:

One of the supervised learning techniques typically utilized in classification problems is the use of support vector machines. On a plane, it is a method for separating classes at a target. A straight-line divides these more than two classes from each other. The objective is to reach the maximum distance between the points of this line's two classes. It generally works well for working with small-scale and complex datasets.

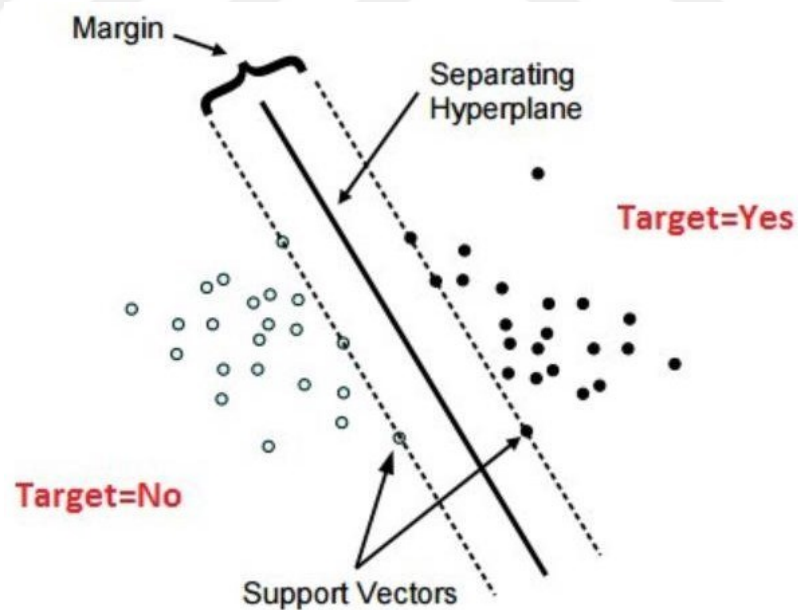


Figure 3.6 Support Vector Machine Algorithm

There are two different classes, black and white, on the table. Our main purpose in classification problems is to give the class of future data. A line that clearly separates these two classes serves as the basis for this classification. This is how the margins are made. The better the separation of more than two or more classes is, the wider the margin.

$$\hat{y} = \begin{cases} 0 & \text{if } \mathbf{w}^T \cdot \mathbf{x} + b < 0, \\ 1 & \text{if } \mathbf{w}^T \cdot \mathbf{x} + b \geq 0 \end{cases}$$

Figure 3.7 Adding New Value Criteria on the Support Vector Machine Algorithm (AKCA, 2020)

Depending on the variance variety, there could be some differences between the classes. A new value will be closer to the white dots if the result is less than zero. On the other hand, the result will be closer to the black dots if it is greater than or equal to zero (Suthaharan et al., 2016).

Try setting the margin in two different ways. Sometimes the margin doesn't work out as planned. Samples may occasionally enter the margin region. Soft margin is the name of this phenomenon. If the data can be separated linearly, it works, and it is extremely sensitive to outliers. Hard margin is the name of this procedure.

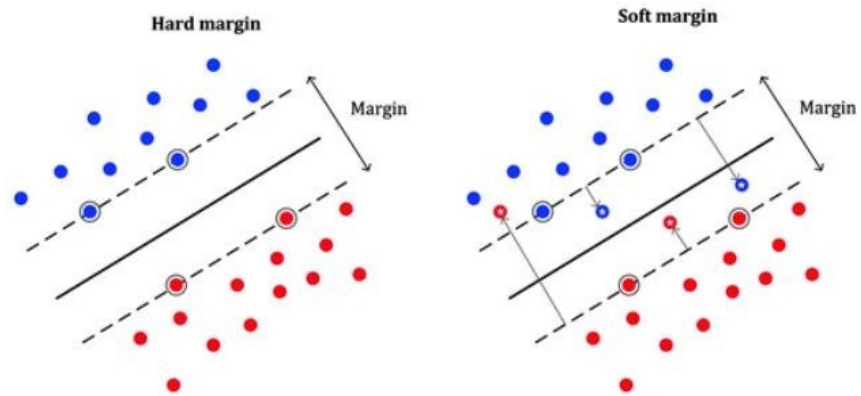


Figure 3.8 Hard Margin and Soft Margin Differences (AKCA, 2020)

The hyper parameter C sets the balance between these two margin methods. The smaller the C , the smaller the margin. The margin is larger the lower the C value. The C hyperparameter ought to be decreased in the event that the model is overfitted.

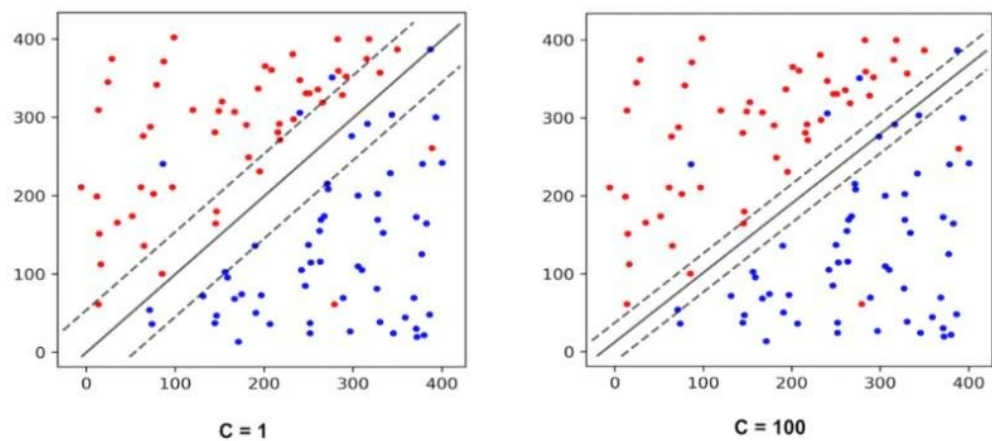


Figure 3.9 Choosing the Best Hyperparameter of Penalty (AKCA, 2020)

For low-dimensional complex datasets, margin adjustments and hyperparameter selections may not be sufficient. The process will take longer than anticipated if the size is increased because there will be more transactions. In this instance, the kernel trick method ought to be used. Certain Kernel Functions can multiply the kernel trick coordinates, giving them a lot more meaning (Pisner, 2020).

- ***Polynomial Kernel:***

- In this approach, the problem is approached as though it were operating in three or more dimensions rather than just two. A straight line cannot be used to classify two-dimensional distributions. For this, we make use of the polynomial kernel. In order to distinguish the classes from one another when working in the third dimension, it is necessary to employ a plane rather than a straight line. The classes are separated from one another more smoothly for this operation.
- If the model is overfitting, its degree is reduced, if it is underfitting, it is necessary to increase its degree. It is also possible to adjust how much the model will be affected by high-order equations, thanks to the `coef()` hyperparameter. This process only happens in poly and sigmoid kernels.

- ***Gaussian RBF (Radial Basis Function) Kernel:***

- It classifies Support Vector Machines of infinite size based on how similar each point is to a particular point using a normal distribution and finds Support Vector Machines of infinite size. The gamma hyperparameter is used to measure the distribution's width. The distribution is wider the smaller the gamma. The distribution becomes more narrow the larger the gamma. Similar to the C hyperparameter, the gamma value should be decreased if the model is overfitting, while the gamma value should be increased if the model is underfitting. If the dataset isn't too big, generally speaking, the RBF Kernel should be used.

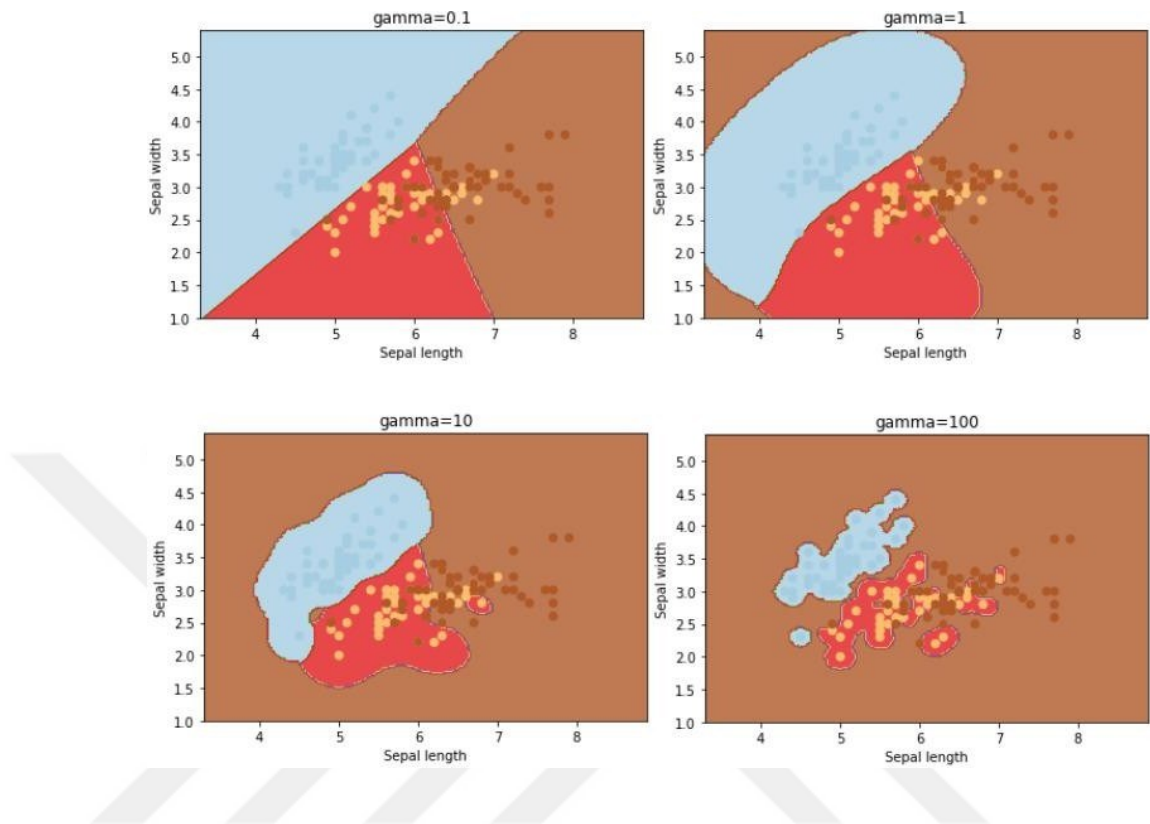


Figure 3.10 Gamma Value Selection Changes on Data Classification (AKCA, 2020)

SVM is a supervised learning algorithm that can be used to separate two or more classes based on their distance from each other with a margin line created by applying the kernel trick method. In this study, SVM was one of the algorithms utilized in working with clinical data. The hyperparameters, including tuning, gamma, and kernel parameters, were tuned using a predefined set of values, and the best parameter values were selected before the model was trained. The optimal hyperparameter values for this algorithm were determined as follows: a tuning parameter of 1000, a gamma value of 0.01, and the RBF (Radial Basis Function) kernel trick method. SVM was executed and based on the accuracy results, it was found that the SVM algorithm achieved more successful results compared to logistic regression. The accuracy of disease classification obtained using SVM was 84.21%.

Decision Tree:

Decision trees are classification algorithms from the supervised machine learning family. Regression and classification problems can both be solved with the help of decision tree algorithms. Using a decision tree, simple decisions from training data are used to train and predict the classes or particular values of the target variable. The algorithm in a decision tree must begin its search at the root based on the predictions of a class or multiple classes for each record. The values of the root attribute need to be compared to the attributes of each record. The branch must be followed all the way to the value using these comparison operations and travel over the branches to all the next nodes (Kotsiantis, 2013).

By dividing the main set into subsets based on a target value condition, the tree structure can be trained. Iterative partitioning refers to the repeated execution of this procedure on each derived subset. It is finished when every subset in a node has the same property of the target variable, or when the split doesn't change the estimates that are left. The decision tree is created without the need for any additional settings, field information, or parameters. As a result, it's ideal for quick

and easy knowledge discovery. According to the study A. D., (2020), the inductive decision tree model yields excellent results for large-scale data.

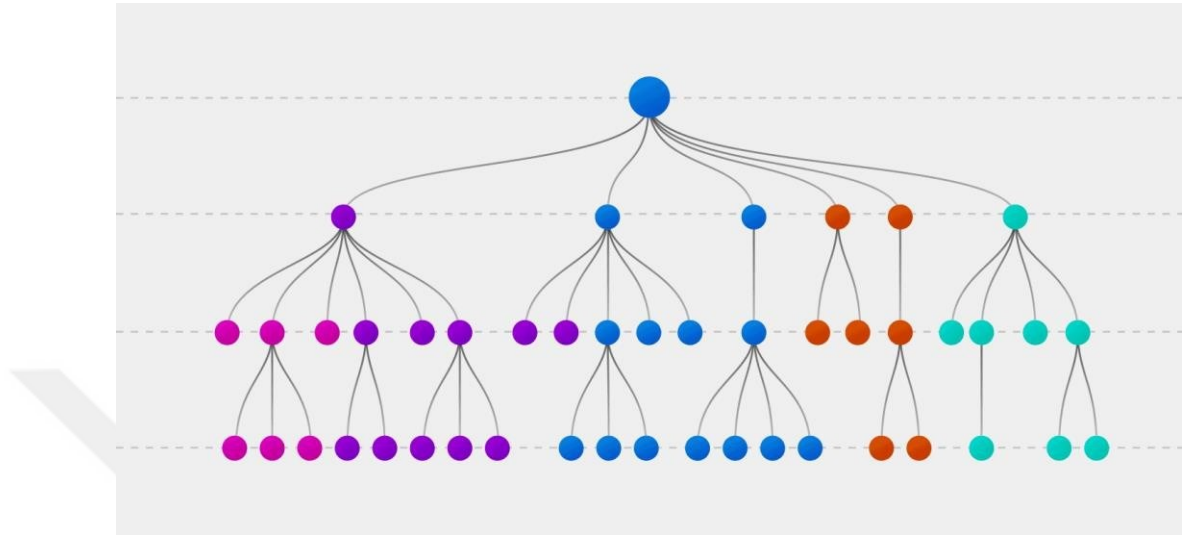


Figure 3.11 An Example of Decision Tree Structures (A.D.,2020)

There are two main types of decision trees. The types of target variables are the source of these variations. The decision tree is referred to as a "Categorical Variable Decision Tree" when the target variable has a categorical value. The decision tree is referred to as a "Continuous Variable Decision Tree" when the target variable has a constant numerical value.

1. ***Categorical Variable Decision Tree***: A decision tree with a categorical variable is used to begin the classification process in the dataset if there is a target value and a categorical target variable.
2. ***Continuous Variable Decision Tree***: A decision tree with a continuous variable is used to begin the classification process in the dataset if there is a target value and a continuous target variable.

The sample values in a decision tree are sorted by their classification from the root to the leaf nodes. The sample value can now be more easily categorized in this manner. An illustration tests the attribute using the data it receives from the root node of the tree before moving on to the leaf nodes. For the subtree that is rooted in the new node, this process then begins to repeat itself. Entropy always comes and goes between this [0-1] interval in this root-node structure. According to Lyman, entropy is not merely a measurement of disorder or purity. It is the measurement of the data points' randomness or impurity. High-level disorders are triggered by low-level impurities. This could be more than one group or class in the datasets, but it still indicates the same disorder issues. Entropy should have been controlled in decision trees to understand information gain and Gini impurity. Despite meeting the rate at which ML models reduce uncertainty, entropy is a degree of randomness or uncertainty. As a high-level disorder, the approximation of the model accuracy to the 1 decrease as entropy rises (Maimon et al., 2014).

Information Gain can be calculated in part by using the concept of entropy. When the idea of entropy comes into play, information gain is used to determine which feature provides the most information about classification techniques based on this idea. In most cases, it focuses on reducing the amount of entropy at the root by measuring the degree of disorder or impurity. A formula for information gain can be obtained by multiplying all base-two probabilities of that class's probability (Tyagi, N., 2020).

$$\text{Entropy} = -\sum_{i=1}^n p_i * \text{Log}_2(p_i)$$

Entropy Formula

Figure 3.12 Entropy Formula

Gini index is another name for the Gini impurity. The number of probabilities that a particular feature will be incorrectly classified when selected at random can be calculated with this assistance. The probability quantities of certain features derived from incorrect selections are calculated by making random selections. Pureness refers to the condition in which all items are linked to a single class over others. The Gini index criterion must be understood in terms of entropy's properties. The Gini index, like entropy, also has values from 0 to 1. The true purity value is zero. In this setting, every element belongs to and is bound to a specific class. The random distribution among multiple classes is represented by a value of 1. A Gini index value of 0.5 indicates that some data have an even distribution. As shown in the figure below, when calculating a decision tree, the features with the lowest Gini index are preferred. The Lorenz Curve is the key to understanding the cumulative possibility functions with values above 0.5 (Priyam et al., 2013).

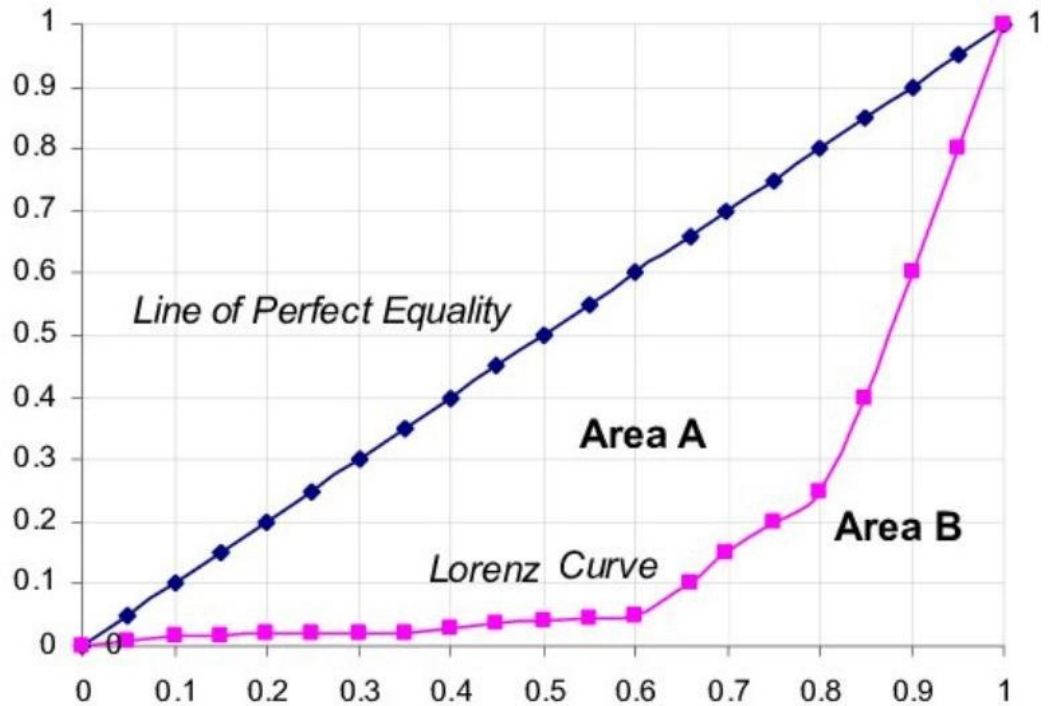


Figure 3.13 Calculating Gini Index with Lorenz Curve (Pitoura, T. and Triantafillou, P., 2008)

In this thesis, the decision tree algorithm is also among the algorithms applied. Thanks to the decision trees, we can reach the final result from the parts called leaves by jumping to the nodes on the tree structure by moving over the binary values of a feature with the calculated mathematical metrics to classify the target feature. At the same time, thanks to the Gini index calculation, the probability of the estimated result on the decision tree is calculated statistically and the result is obtained by calculating the probabilities of more than one feature. Before inserting the train dataset into the Decision Tree algorithm in the kernel coded with Python, hyperparameter optimization was performed and the best parameters were calculated, and the accuracy rate was recorded after the model training. The best hyperparameter and its value can be said as follows. Maximum depth is set to 1. Decision tree was run and according to the accuracy results, the Decision algorithm has achieved nearly similar successful results as Support Vector Machines. The disease classification results obtained as a result of the decision tree were observed with an accuracy of 81.57%.

Random Forest:

The classifier machine learning algorithm known as random forest encapsulates and executes multiple decision trees and is widely utilized in data science. Regression and classification problems can both benefit from this machine learning model.

Before using the random forest algorithm to train the dataset, three hyperparameters must be adjusted. The number of features that need to be sampled, the number of trees, and the size of the node are all examples of this. The random forest algorithm is used to solve regression and classification problems thanks to these parameters.

A collector algorithm with multiple decision trees is the random forest algorithm. Here there are structures with multiple variations in the dataset with multiple features. Another randomness is grafted into the model through feature collection, adding more diversity and reducing the correlation between black trees. Decision trees collected individually in the collection are average, for a classification operation the most common variable or the class estimated over the value with the majority vote is shown. Finally, after the sample training, cross-validation is applied and prediction is made, and test results are obtained (Ali, 2012).

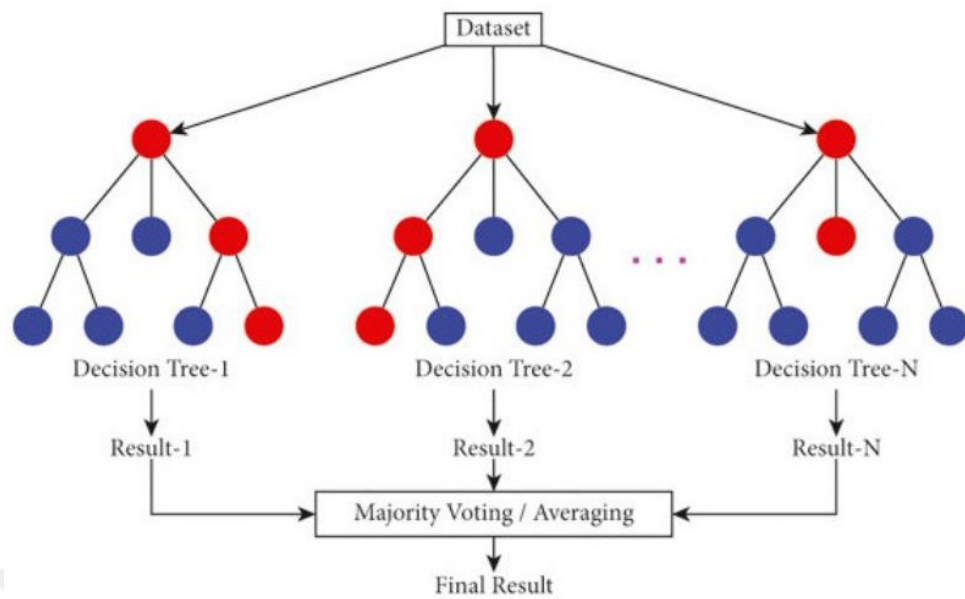


Figure 3.14 An Example of Random Forest Structure (Khan et al., 2021)

As with any model, this model has its pros and cons. Some of the benefits can be itemized as follows (Biau, 2016).

Major Benefits;

- Reduced risk of overfitting.
- Provides flexibilities.
- Determines feature importance easily.

Major Difficulties;

- Requires More Resources
- Time Cost

In order to maintain a tight accuracy within the training data, it may risk overfitting at the training level. Averaging the trees that are less related or unrelated

to one another reduces the overall variance because it is a system with multiple decision trees. Accordingly, a good estimation cannot be made, and the real estimation begins to drift away. In this way, the classifier cannot fit the model much.

It is a very popular algorithm today because it is one of the preferred models in both regression and classification problems. Because it implements feature bagging, people working in the field say it's an effective tool in classification as a random forest classifier that can maintain accuracy in some tests because it doesn't take some of the data.

Thanks to this algorithm, it has a structure that makes it easy to add features and contribute to the model to which the training is applied. We can evaluate the importance of the feature in several ways. Increasing or decreasing the Gini purity is a technique generally used to exclude a certain variable and measure how much the accuracy of the model decreases and how much the loss increases. The mean reduction method also triggers the known permutation significance. By granting random permissions, the average drop in accuracy is defined (Probst et al., 2019).

Random forest algorithms are algorithms that make much better predictions for larger datasets. It is a time-consuming method as it is run with large data. The time cost is quite high. At the same time, because they work with more data, they need more resources to store this big data. The comparison of random forests is more complicated when looking at a single decision tree because it contains decision trees that are more than one variety and whose relevance ranges from 0% to 100%.

As explained above, it is an algorithm that contains more than one decision tree and makes mathematical calculations in the background, considering the results

obtained from each of them, and is used to classify each observation value. This algorithm was tried in this study in the hope of giving better results than the decision tree, but no significant difference was found. With this algorithm, before the data is run, the hyperparameter definitions are defined as in other algorithms, the parameters that give the best results and their values are recorded. The number of best estimators was chosen as 6, the number of features 7, and the max depth 7, and the accuracy rate was recorded as 81.57% ratio.

Despite these benefits and difficulties, the random forest algorithms applied to the clinical data in this thesis achieved some high results compared to logistic regression machine algorithms. It can be proved by the thesis results that the accuracy rate is based on 81.57%.

AdaBoost:

In 1997, the Boosting algorithm's foundations were established. Since then, boosting has become a common method for resolving issues with binary classification. It aims to train the model by converting values from two distinct sample groups to one another with the help of these algorithms. The fundamental principle of this algorithm is to first build a model from the training dataset and then give that model another chance to fix errors. These procedures continue until the rate of prediction accuracy increases and errors are reduced to a minimum. To achieve the desired result, strengthening algorithms aim to combine multiple models with weak or strong classes (Schapire, 2013).

Boosting algorithms can be broken down into three categories. AdaBoost, Gradient Descent, and Xtreme Gradient Descent are the names of these algorithms. The AdaBoost algorithm was the subject of research within the scope of this thesis. One of the most widely used classification algorithms is AdaBoost, which uses adaptive boosting to create single-level decision trees. On the training set, this algorithm creates a base model. All data points are given equal weights. Even higher

weights are then assigned to the misclassified points. Higher weight points are more important in the second model. These are the points that are tried to be strengthened from weak clusters. This training continues until you get an error on the model (Shahraki et al., 2010).

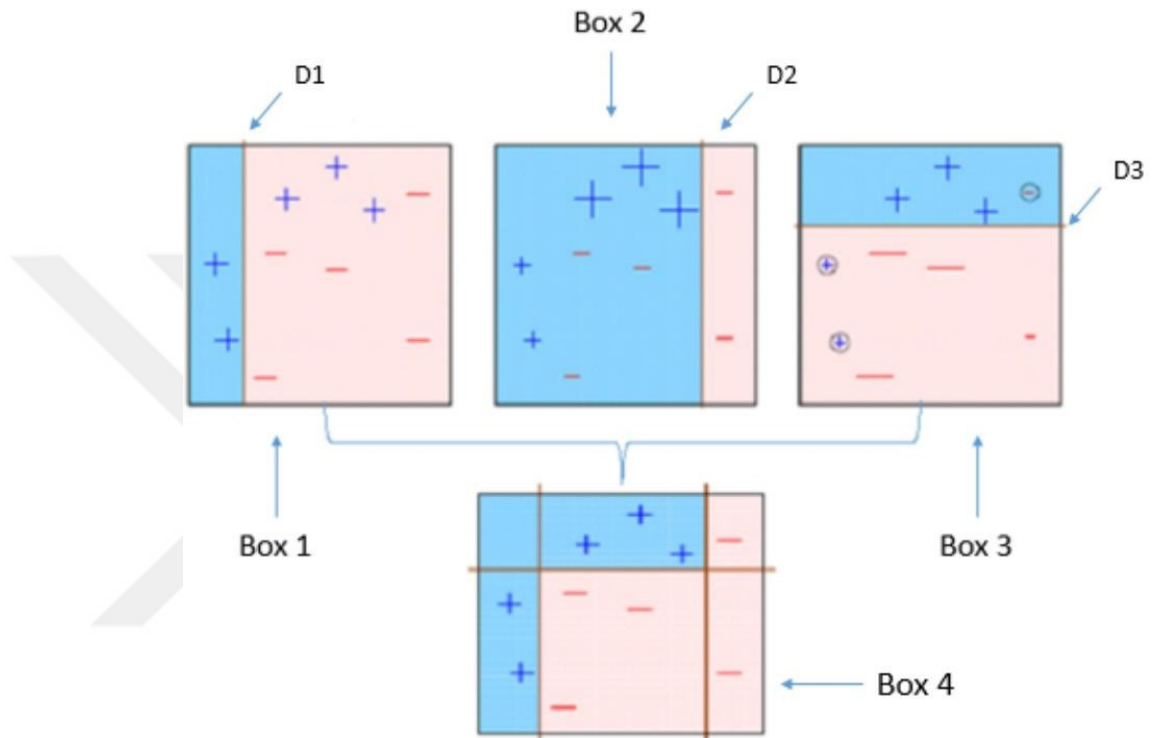


Figure 3.15 An Example of AdaBoosting Algorithm (Akash, 2019)

Boosting algorithms are actually powerful algorithms that can strengthen weak features and positively guide the future of classification. Boosting algorithms are algorithms that can give very good results in diagnosis in the field of health. For this purpose, adaboosting algorithm was also studied in the thesis study. The Adaboosting algorithm ranks first as the algorithm with the best accuracy when working on clinical data. The number of predictors is six, the learning rate is one as the hyperparameters giving the best accuracy and their values.

With these enhancement techniques, results similar to the obtained accuracy in random forest algorithms have been recorded. It can be said that this is the best algorithm for the dataset. However, since the recall value is not closer to 1 as in the random ratio, it can take the title of the best algorithm after random forest based on the dataset working within the scope of this thesis. The accuracy of the AdaBoost algorithm in this dataset is recorded as 86.84%.

On the other hand, In the method applied for ADNI data, 3D convolutional neural network techniques were used twice. To increase the accuracy values and help to reduce the loss, the early stopping criterion was created, and the results were obtained by using the SoftMax layer, which helps to achieve higher results in multi-class studies to take the class decision.

RNN:

Computer systems based on artificial neural networks are constantly being developed to automatically realize the capabilities of the human brain, such as the ability to derive new information and create and discover new information through learning, without the need for human intervention. An artificial neural network that learns from previous experiences is called a recurrent neural network. This could be considered a subset of an artificial neural network.

Models were created by imitating this learning process in mathematical ways by taking the learning methodology in the human brain as an example. Systems have been developed that imitate the features of biological neural networks in the human brain, such as learning, remembering, understanding, generalizing, or extracting parts from the whole. These learning processes are taught to the machine using examples. Rules are defined on the input and output information given during learning, and artificial neural networks are created by following an approach close to neural networks in the human brain. The results of the trained machine learning models are recorded and further developed.

Artificial neural networks are constantly developing computer systems that can automatically recognize the capabilities of the natural human brain, such as taking in new information, creating, and discovering new information through learning, through machines without any human assistance. A recurrent neural network is an artificial neural network that learns from previous experience. It can be considered as a subset of artificial neural networks (Graves, 2013).

Artificial neural networks consist of neurons. Neurons are units that can process information. As in our brain, neurons connect and create functions. It is known that there are more than one hundred billion neurons in today's human brain. A neuron can connect with another neuron at least 50,000 and at most 250,000. In this way, we can follow, examine, and understand the behavior of countless living things and interpret them with output information. The production of artificial models similar to the structure of this human brain is called cybernetics. Relevant units can be trained on functions to achieve the desired output. Although artificial neural networks are algorithm-based systems, in some cases they may not follow a definite method to reach the desired or unwanted information. This can be shown as the biggest proof that they do not have a fixed algorithm (Von Bartheld et al., 2016).

Before artificial neural networks can be understood, it is necessary to make sure that one understands how a nerve cell in the human brain talks to other cells.

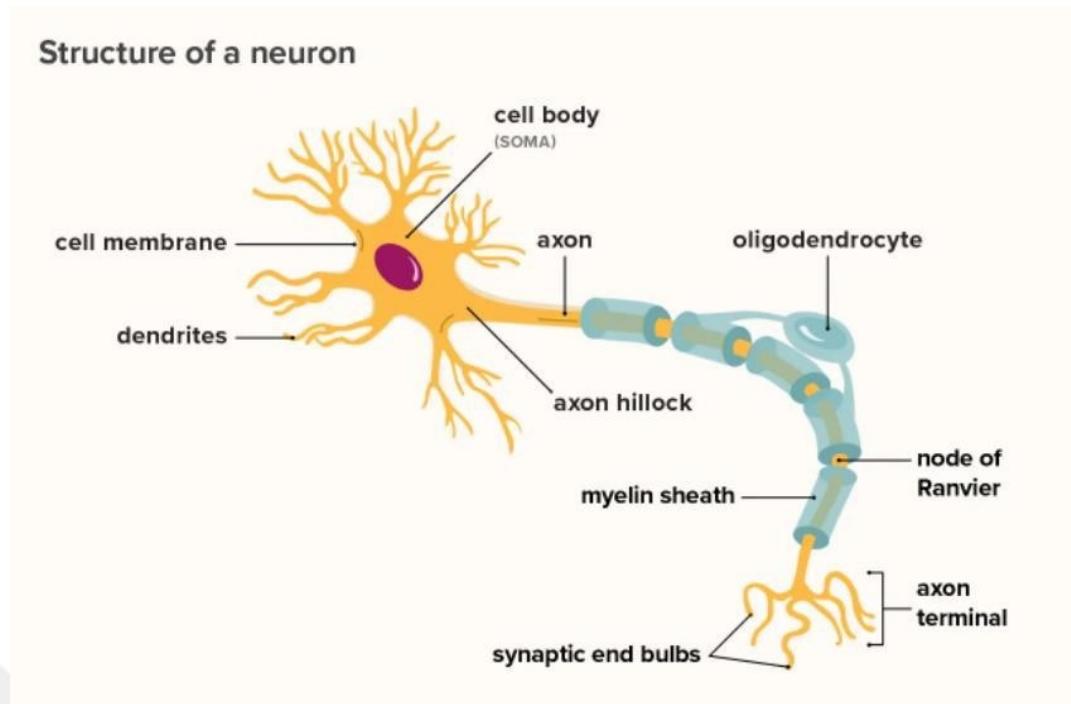


Figure 3.16 Real Human Brain Neuron Parts (Vandergriendt, 2022)

A nerve cell conducts electrical transmission to the nucleus via dsajip and axons. Output terminals contain synaptic sensors located on the dendrites. Thanks to these synaptic arms, information is transmitted to another nerve cell (Kusumoto, 2019).

The mathematical model of a human neuron can be summarized in the model below.

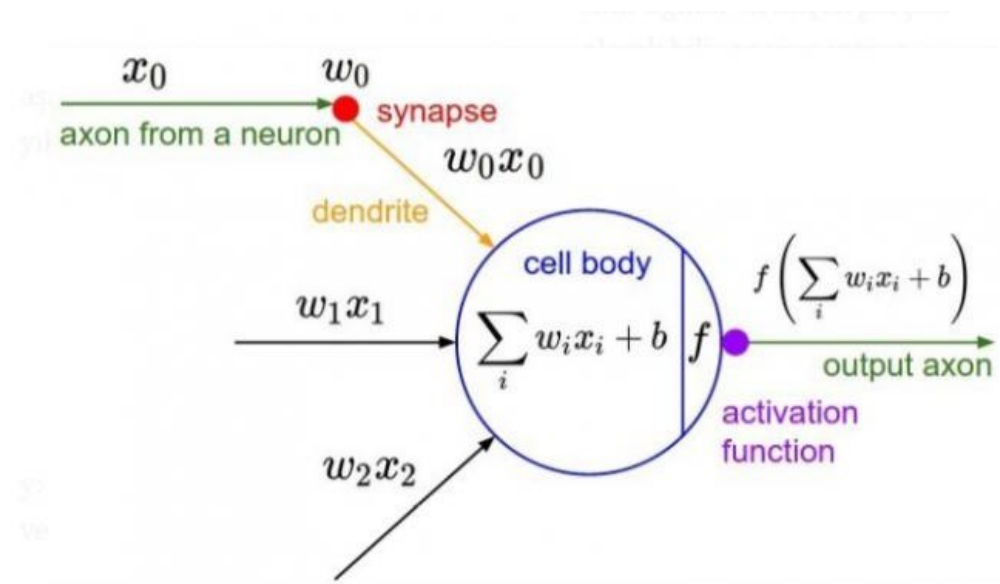


Figure 3.17 Artificial Neural Study Designed Based on the Human Brain

- x_0 7 An input from across dendrites or from another neuron.
- w_0 7 The weight in dendrites.
- $w_0 x_0$ 7 The value obtained by multiplying the weight in the dendrite by the unit information. This calculation happens inside the neuron.

After that, all of the multiplications are added together to determine the total weight of the information. After a bias value has been added to the weighted sum, this is the overall result's weighted sum. The output is ready to display after the activation function. This output could either be the end result or an input from a different neuron cell. The model's best score is determined by calculating the parameters known as weight parameter and bias value which is called as B value. To summarize, the artificial neural network includes five main pieces (Squire et al., 2011).

1. **Inputs:** Neurons receive information from inputs. Like in biological nerve cells, data from these inputs are sent to the nucleus of the neuron for collection.
2. **Weights:** Before reaching the nucleus through the inputs, the information that enters the artificial nerve cell is multiplied by the weight of the connections from which it originates. Adjusting the impact of the inputs on the final output is possible in this manner.
3. **Addition Function (Combination Function):** By multiplying the inputs by weights, the addition function calculates the net input of an artificial neuron cell.
4. **Activation function:** A function that generates a non-linear output value from the weighted sum of all the inputs in the layer before passing it on to the next one. (e.g., SoftMax, ReLu, or sigmoid).
5. **Outputs:** The cell's output value is the result of the activation function's output. Despite having multiple inputs, each cell only has one output. Any number of cells can be connected to this output.

There are three primary levels of examination for artificial neural networks. The output layer, the hidden layer, and the input layer are these (Keren et al., 2016).

The input layer is where data from the input is sent. The input data is processed and sent to the final output layer in the middle-hidden layers. The operations in the middle layers are meant to turn into output the weighted sum of the product of the input information and the network's weight values. The weights need to have the right values in order for the network to produce the right outputs.

As seen *Figure 3.18*, if the model consists of many intermediate layers and contains many neurons, this model is called a multilayer neural network. If not, it is defined as a single-layer neural network.

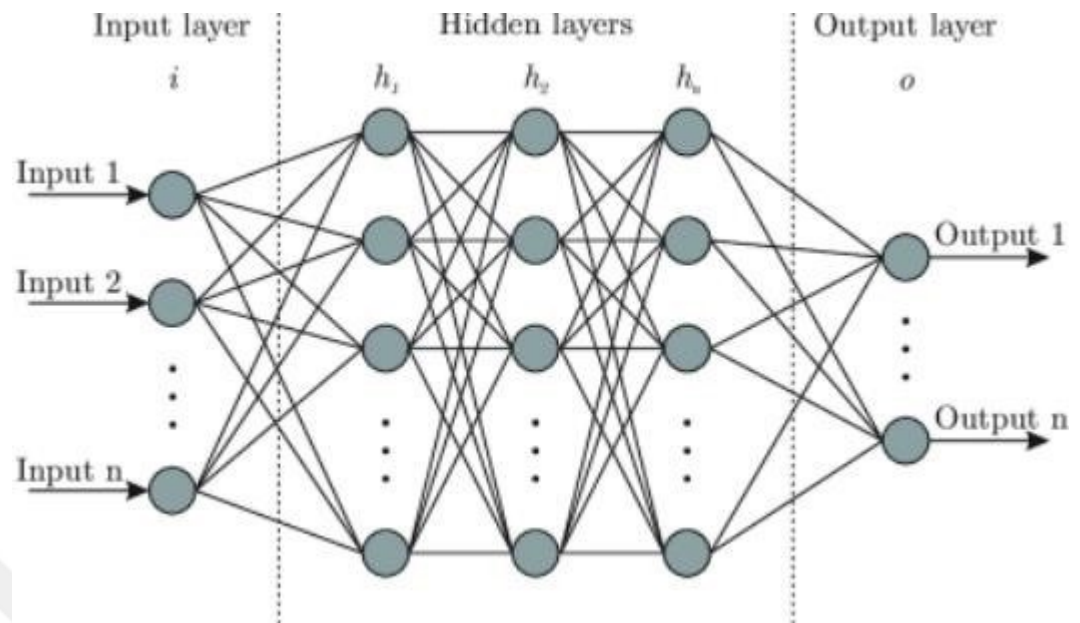


Figure 3.18 An Example of Artificial Neural Network (Bayhan, 2022)

- **Single-layer Neural Networks:** The neural network's input can either be added directly to the output or added at the following layer. Typically, they are two-input, one-output networks. Using their weight multiplied by a threshold value, inputs travel along synapses.
- **Multi-layer Neural Networks:** Therefore, it is a neural network with many inputs that have multiple result values in more than one intermediate layer but contain a final basic result in the output layer, which is preferred in nonlinear problems.

In this thesis, data in image sets are divided into four different groups;

- Mild Dementia
- Moderate Dementia

- Non-Dementia
- Very Mild Dementia

When working with image sets, the primary objective is to perform numerical encoding using the One Hot Label Encoding method. This enables the classification of the qualitative disease feature by assigning grades to the target feature using this encoding. The training set is segmented into train and validation sets, with RNN layers configured on the fragmented training set. These RNN layers consist of five layers with activation functions, four of which utilize the ReLu method, and one which utilizes the SoftMax metric. SoftMax is preferred for its ability to perform classification in a target variable that selects more than one class with greater accuracy.

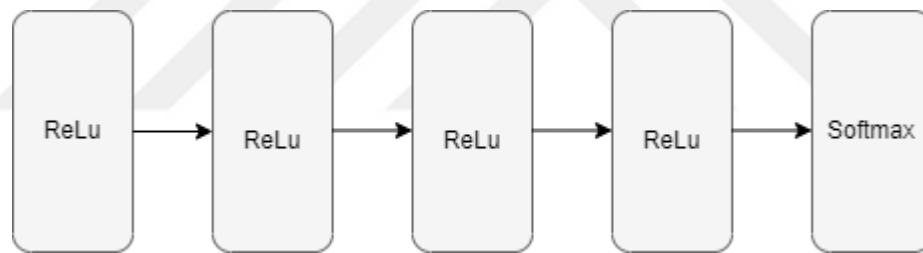


Figure 3.19 Activation Function Orders

In this study, the one hot label encoding method was employed for image classification on a dataset comprising four distinct classes. The dataset was partitioned into train-test-validation sets, and a prefetch process was applied to enhance latency and output results. To prevent overfitting, the early stopping criteria was established and the model was defined and trained with the Adam optimizer. The model was compiled with 50 epochs returned for RNN, and the performance of the model was evaluated with the test dataset. The results indicated an accuracy of 77.87% and model loss of 1.039, demonstrating the effectiveness of the proposed approach.

CNN:

When compared to other types of neural networks, convolutional neural networks are renowned for their superior performance on audio and image data. The main layers are three. These are;

- Convolutional layer.
- Pooling layer.
- Fully connected layer.

When the CNN is set up, the second layer that is created is called the convolutional layer. The final layer is the fully connected layer, while additional convolution or pool layers can follow the convolution and convolution layers. The pooling layer in this case is comparable to the hidden layers we set up in ANN. Through operations at each layer, CNN tries to increase complexity by identifying large parts of the data. The previous layers focus on detailed features. As the image or audio data progresses through the CNN layer, it tries to understand the larger shapes, different structures, and details of that element until it finally parses and understands the object it will be describing (Kattenborn et al., 2021).

The convolutional layer in this structure is the base constructed block in a convolutional neural network. This is the part where the actual calculations take place before all the operations in the middleware.

Convolutional Neural Network takes an image as input information. The image information is separated into three color planes in RGB Codes (Red-Green-Blue

Codes) and detects the multiple color spaces inside of the image information which has been given as input image to the CNN. It also measures the image dimensions.

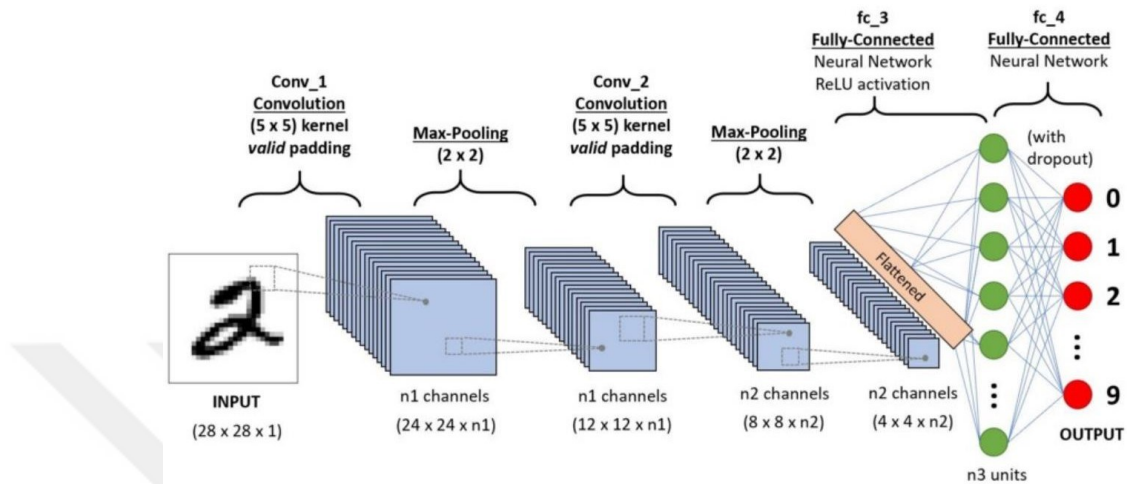


Figure 3.20 A Sample of CNN Workflow (Aigents.co)

In an image, there is three color plane which is red, green, and blue codes. They are also defined as CMYK or Grayscale, etc...

This process is to work on the color codes, which can be used to make a better estimation. Even the most detailed information here affects the neural network so much that these architectures that we run on very large image sets have a very important place.

CNN has the formula to convolve the unit picture.

Image Dimensions = Height * Breadth * Number of Channels (RGB Codes)

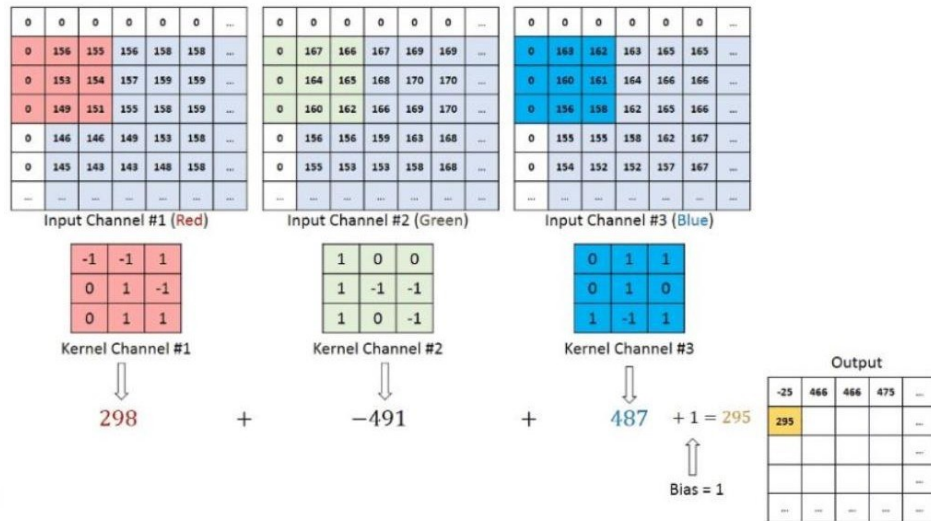


Figure 3.21 Convolutional Sample Calculations in Neural Network (Aigents.co)

$n \times n \times 1$ dimension is convolved to the $m \times m \times 1$ image shape and the colorful sides move to the non-traced places for the value of strides until all the image block is completed one by one for the convolutional neural network model. This parses the image until it finishes the breadth.

It works at the pooling layer in the direction of reducing the greatness of the rotated feature. It works to reduce the computational strength for processing data alongside size reduction operations. It is also useful for extracting master features and dominant properties that are rotationally and positionally invariant. Therefore, the progress must be maintained properly (Li et al., 2021).

In literature, there are only two kinds of pooling techniques;

- **Max Pooling:** Supports the max value inside of the kernel.
- **Average Pooling:** Supports the average value inside of kernel.

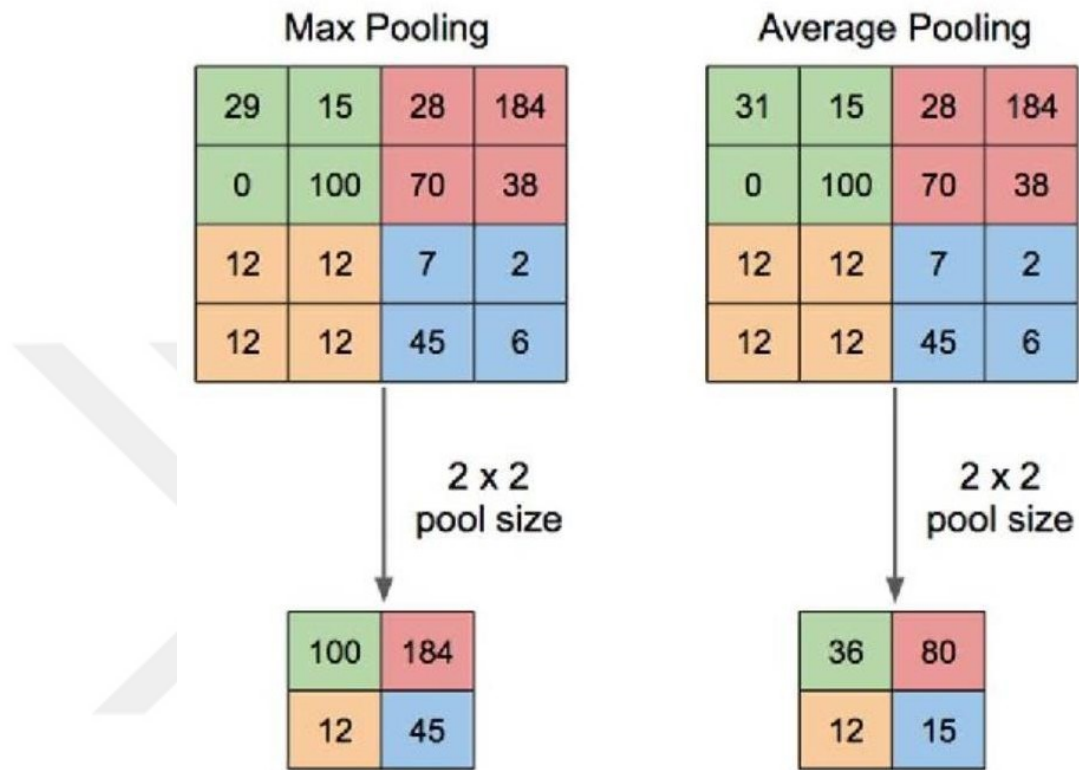


Figure 3.22 Pooling Methodologies (Yani et al., 2019, p. 3)

Values with non-linear combinations out of concrete context are said to be the easiest way to learn the outcomes obtained as a result of learning. Allows space to learn nonlinear functions. The output of the image layer is translated into multiple sensor forms. Some CNN architectures can be itemized as (Wu, 2017);

- AlexNet
- GoogleLeNet
- ZFNet

- LeNet
- ResNet

In this thesis, data in image sets are divided into 4 different groups as;

- Very Mild Dementia
- Mild Dementia
- Moderate Dementia
- Non-Dementia

While working on image sets, it is aimed to do digital encoding with the One Hot Label Encoding process. By using this coding, our target feature is graded and our disease feature, which is defined as qualitative data, is classified. The training set is divided into two basic parts as train and validation. CNN layers are constructed on the fragmented training set. In these layers, one MaxPooling operation is performed after two basic layers. Normalization and MaxPooling process continue after defining convolutional layer three times and two times separately. The transformation used in batch normalization keeps the standard deviation of the output close to one and the mean output close to zero. In CovNet as H. et al, (2019) described in their study, this layer is frequently added between convolutional layers that come before it. This layer's task is to reduce the number of network parameters and calculations as well as the size of the representation's displacement. In this manner, the network's incompatibility is checked. Layer dropout is a technique to reduce overfitting on the model. These are repeated one more time and the output is flattened to a one-dimensional array format. With the Dense Block process, normalization and dropout are done by adjusting the dense. “Dense” representing layers is one of the most used classes in Keras. It provides the transition of neurons or nodes between layers with dense. In this architecture, neurons formed in one layer are transmitted to another layer by weighted

calculations. Thus, communication is established. Model accuracy is observed at almost 81.34% with the help of the CNN Structure that is implemented as shown in *Figure 3.23*.

Similar operations have been implemented in CNN, but with different values. In our data set, which is divided into four basic classes, firstly, one hot label encoding method was applied to the images, and then they were visualized according to their classes. Train – test - validation was divided into parts and prefetch process was applied. In his process, the work of the current running steps and the preparation of the next steps are obtained and improved latency and output results.

Afterwards, the early stopping criteria was determined, and the parameter was defined to prevent the overfitting process, the CNN model was defined with the help of TensorFlow libraries, and the training process was started. In this study, 100 epochs were returned for CNN. The value obtained as a result of 100 epochs was evaluated with test data sets and it was observed that the result was 81.34%. In this research, it has been recorded as an artificial neural network that works with image sets and gives the best results.

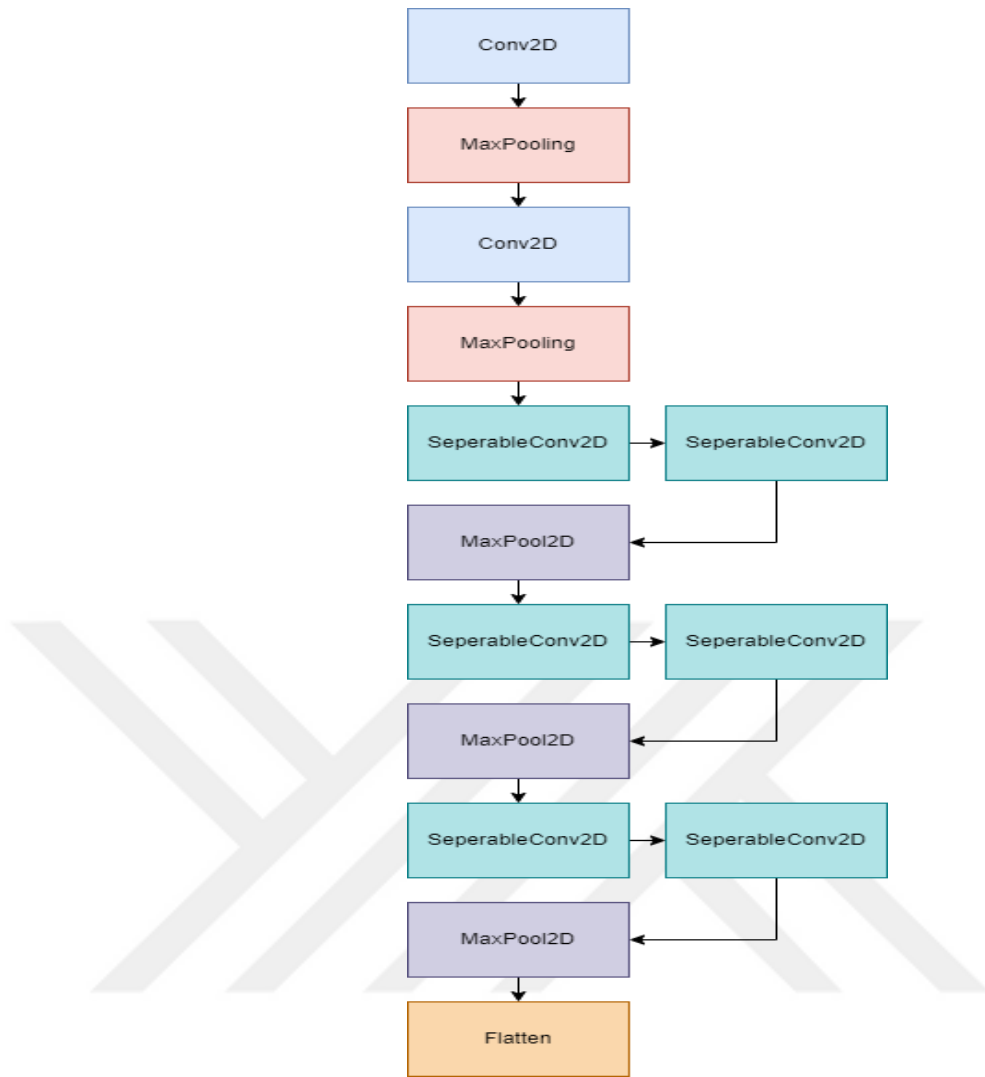


Figure 3.23 CNN Structure of Recommended System

VGG19:

VGG is another convolutional neural network. This architecture is used to increase the depth of the weighted networks' layers. The network focuses on 3*3 filters mainly. It has been announced to the public as one of the algorithms that allow us to achieve much better results for computer vision processing rather than image classification (Carvalho et al., 2017).

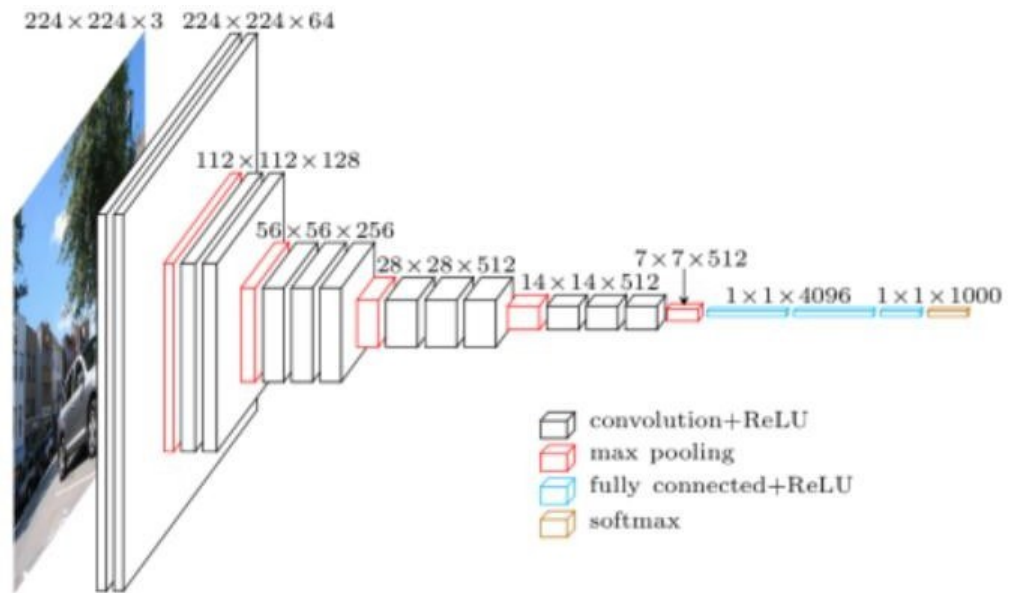


Figure 3.24 A Sample of VGG Workflow (Papers with Code – VGG Explained)

VGG is a family of deep convolutional neural networks that have been trained on large-scale image classification tasks. They are widely used in various computer vision applications, including image recognition, object detection, and segmentation. The VGG network architecture was introduced in 2014 by a group of researchers from the University of Oxford, and it achieved state-of-the-art performance on the ImageNet Large Scale Visual Recognition Challenge

The main characteristic of the VGG network architecture is its depth, which means it has many layers. The original VGG network consisted of 19 layers, including 16 convolutional layers and 3 fully connected layers. The convolutional layers are responsible for learning the features of the input images, while the fully connected layers are responsible for making the final predictions. VGG networks use small filters (3×3) with a stride of 1, which enables them to learn finer details of the images.

VGG networks have also introduced the concept of transfer learning, which refers to the use of pre-trained models for new tasks. The idea behind transfer learning is to use the learned features of a pre-trained model on a large dataset to improve the performance of a model on a smaller dataset with fewer examples. This is particularly useful in cases where there is limited data available for a new task. By using a pre-trained model as a starting point, the model can leverage the knowledge of the pre-trained model to learn more efficiently and effectively.

Overall, VGG networks have proven to be effective in a wide range of computer vision tasks and have been widely adopted in the research community and in industry. Their deep architecture and use of transfer learning have contributed to their success and continue to inspire new developments in the field of computer vision. These are some benefits of VGG structure (Yadav et al., 2023)

- High accuracy
- Easy to understand
- Transfer learning
- Interpretability
- Open-source implementation

In this thesis, data in image sets are divided into 4 different groups as

- Very Mild Dementia
- Mild Dementia
- Moderate Dementia
- Non-Dementia

While working on image sets, it is aimed to do digital encoding with the One Hot Label Encoding process. By using this coding, our target feature is graded and our disease feature, which is defined as qualitative data, is classified. The training set is divided into two basic parts as train and validation. The training folder as training and validation is fragmented. The aim here was to increase the quality of training by validating the model while training it with the training set. After evaluating the results with the test, it was better understood that the model learned the most. Before entering the testing process, it was aimed to train 50 epoch models, but before starting this training, it was allowed to define hyperparameters in the model for improvement against memorization problem. Early stopping methods have been added to minimize the overfitting problem. In this way, the accuracy rate in the VGG architecture has gone even better, leaving the initial accuracy rates behind.

The same basic steps applied in RNN and CNN were also applied for VGG (Very Convolutional Neural Network). In our data set, which is divided into four basic classes, firstly, one hot label encoding method was applied to the images, and then they were visualized according to their classes. Train – test - validation was divided into parts and prefetch process was applied and improved the outputs that are provided from the structure model.

Afterwards, the early stopping criteria was determined, and the parameter was defined to prevent the overfitting process, the CNN model was defined with the help of TensorFlow library, and the training process was started. In this study, 50 epochs were returned for VGG19. And after so many epochs, the resulting value was evaluated with test datasets, and it was observed that the result was 77.49%.

4. RESULTS

First of all, the results of the studies are discussed, and then, before working on this thesis in the literature, we will proceed by comparing the results of the thesis with the researchers that were considered competitors at the time of the research.

In this thesis study, both clinical and image data were researched, the accuracy rates on the datasets operated with different algorithms and the relevant metrics were noted, and the results were recorded by establishing a cause-effect relationship between the models.

In the study conducted on clinical data, five basic classifier algorithms were studied. The algorithms which are used in clinical data for this research are logistic regression, random forest, support vector machines, decision tree and adaboosting. It is said that the third-highest accuracy rate was obtained in random forest with the highest recall and AUC values, when the accuracy values obtained were compared among the model techniques used. Unlike the single-tree algorithm, Random Forest is known as a multi-tree algorithm, so it is likely to get better results than the decision tree. AdaBoost is a boosting algorithm, and it is chosen as the best algorithm for this tabular dataset. In this research, AdaBoost gave the best accuracy ratio with spontaneous recall and AUC values. If we choose the second algorithm when adding into the recall and AUC metrics, this would be the Support Vector Machines, this is the balanced accuracy results when comparing with other output metrics.

	Model	Accuracy	Recall	AUC
0	Logistic Regression With Imputation	0.763158	0.70	0.70
1	Logistic Regression With Drop Blank Values	0.789474	0.70	0.70
2	Support Vector Machines	0.842105	0.70	0.70
3	Decision Tree	0.815789	0.65	0.65
4	Random Forest	0.815789	0.75	0.75
5	AdaBoost Algorithm	0.868421	0.65	0.65

Figure 4 1 The Overall Result of Clinical Data

Both ANN, VGG, and CNN artificial networks were designed on the image sets taken over ADNI and the results were observed. It is said that the most successful neural network here is CNN. CNN has achieved a rate of 81.34 %, leaving VGG and RNN behind by 3%.

Table 1 Comparison Between the Suggested Researches and Target Researches

Year	Authors	Dataset	Validation	Machine Learning	Performance (Accuracy)
2007	Kun et al, 2007	44(18AD, 26CN)	-	LR	81.1%
2009	Yong et al., 2008	98 (37 CN, 37 AD, 12 FTD, 12 CN)	LOOCV	PC	84.3%
2009	Chris et al., 2009	183 (89 AD, 94 CN)	LMOCV	LP	84%
2011	Anil et al., 2011	129(69 AD, 60 CN)	10-fold	LR	85.25%
2013	Gabriela et al., 2013	1064 (295 AD, 335 CN, 261 sMCI, 173 MCIC)	7-fold	OPLS	84.4%
2015	Siqi et al., 2015	758 (204 CN, 180 AD, 214 MCIC, 160 MCInc)	10-fold	SAE	82.10%
2016	Xiaofeng et al., 2016	202 (52 CN, 43 MCIC vs. 56 MCInc, 51 AD)	-	LDA + LPP	61%
2019	Latha et al.,2019	812 (227 CN, 396 MCI, (165 MCIC,	k-fold k = 1 to 50	SVM + RF + KNN	87.39%

		223 MCInc), 189 AD)			
2019	Hongfei et al., 2019	833 (221 AD, 297 MCI, 315 CN)	10-fold	3D-CNN	93.61%
2022	Recommended Method (Tabular)	373(190 NonDemented,146 Demented, 37 Converted	5-fold	LR, SVM, DT, RF, AdaBoost	76.31 - 78.94%, 84.21%, 81.57%, 81.57%, 86.84%
2022	Recommended Method (Image)	5121(717 MildDemented, 52 ModerateDemented, 2560 NonDemented, 1792 VeryMildDemented)	20-fold	RNN, VGG19, CNN	77.87%, 77.49%, 81.34%

Among the other methods, the model produced by adapting the 3D CNN method to the classification problem of Alzheimer's Disease, with an accuracy performance of 93.61%, obtained the highest accuracy rate. Although the CNN method is followed on the image sets we have, it can be said that the accuracy value remains at 83.67%, due to the difference in data and the 3D CNN model is not followed. In this case, that can be said according to the Latha et al., (2019) research, our model reaches slightly lower accuracy values when compared to competing data. Recommended model in the study of Kun et al. (2007) represents a higher result in the research that did in that year. It is a slightly different between the model's accuracy with the other models' accuracy. The following methods and models seem as closer to each other but the difference can come from the data variety. This also can be said that the accuracy value obtained in the clinical data of this study in the machine learning models applied is due to the redundancy of data in his study.

The method proposed in this thesis has two main arms. The first part is the methods suggested on clinical data. logistic regression, random forest, support vector machines, decision tree and adaboosting algorithms were followed on tabular data.

The second part is the suggested methods for image data. RNN, CNN, and VGG19 methods were followed on the image data. The results obtained are available as of title [3.4. Implemented Methods](#) of the thesis report.

This paper studies a system that can diagnose Alzheimer's disease from an Alzheimer's patient's medical history and assist the physician in the diagnostic process by performing the necessary analysis from the patient's clinical and imaging information. Two of his main studies were done to diagnose this disease. The first arm of these studies was performed using clinical data, while the second arm was performed using imaging data. The clinical data were tabular with patient characteristics representing basic social and medical characteristics, and the image set was the patient's MRI images of her. Although the methods used in these two different studies are fairly independent, the general flow is similar. Clinical data were collected as tabular data from the OASIS database containing patient and non-patient medical records. After exploratory analysis of these data, data preprocessing and correlation analysis were performed. Apply data visualizations to see emerging relationships more clearly, then observe and report results using machine learning classification algorithms such as logistic regression, support vector machines, decision trees, random forests, and Ada Boosting and achieved the desired classification state. Definition of hyperparameters. Image data includes data in classified image format from the ADNI database, aggregated by disease severity. After an exploratory analysis was performed on this data, the dataset was split into training-test-validation, providing data preprocessing steps to determine which images were diagnosed with hot label coding at which disease stage and how many. Some parametric criterion was used to determine minimization and prevent overfitting processes. Finally, there are defined artificial neural networks, created models and layers, trained data using these models, and evaluated results using data from a test set. This tested how well the model learned from the training process and calculated accuracy. These studies have shown that boosting algorithms perform better on clinical data and convolutional neural networks perform better on image set studies. The research conducted in this thesis yielded much better results than some studies in this field and many studies that followed similar methods.

5. DISCUSSION

Today's technology is constantly renewing and evolving. With the acceleration of these developments, studies and practices that will contribute to humanity are being carried out and implemented. The common goal of all researchers and people who are devoted to this field is to improve the living standards of humanity and to reach our next good. Every research done more or less affects the next generation.

Nowadays, it is seen that there are many studies in the field of software, sales-marketing-finance, logistics, retail, smart process automation, pharmaceutical research, and health.

Artificial intelligence used in healthcare follows a broad method used to describe machine learning algorithms, software, and artificial intelligence to mimic human percipience in the analysis of complex medical and healthcare data. It would be the best method to separate the artificial intelligence scientists use in the field of health from the methods used in other fields. Here, traditional patterns are broken. What distinguishes it from traditional methods is its ability to collect data, process data, and give a well-defined great output to the end user. AI (Artificial Intelligence) does this through machine learning algorithms and deep learning (Wikimedia Foundation, Inc., 2021).

Machine learning algorithms and deep learning structures can identify patterns of treatments and similar patterns and reveal their own will. They begin to conclude their logical line with what they have learned. A very large amount of data must be used to reach a useful result. Thanks to the redundancy of data, it is possible to reach better results within the sample.

The first and main purpose of health oriented artificial intelligence applications is to understand and analyze the relationship between forethought or cure techniques and the outcomes, results of patients.

Machine learning techniques are applied for applications such as diagnosis progresses, cure techniques, treatment protocols, drug development, personal treatment methods, patient monitoring, disease diagnosis, early diagnosis, and patient care. Thanks to digital or manually written health records, large amounts of data are analyzed, and great studies are carried out.

In this study, the most important features that we can call the most helpful features in reaching the target in clinical data;

- Years of education
- Socioeconomic status
- Mini-mental state examination
- Clinical dementia rating
- Atlas scaling factor
- Normalize whole brain volume
- Estimated total intracranial volume

Education level and socio-economic status are seen as the two characteristics with the highest logical correlation with each other. It is seen that the education level of a patient is directly proportional to the economic status. It was observed that patients with poor economic status or low education levels have a high probability of catching this disease. People who do not have sufficient financial means are malnourished and their housing conditions are worse than others. These

factors are seen as features that are hidden in this dataset. This situation directly affects their nutrition, shelter, and basic needs.

While the zero-value shown in the legend in the table is the records of the patients without the disease, the data showing as 1 are the records of the individuals with the disease. It is possible to see the breakdown of the education level over the target trait in the table below.

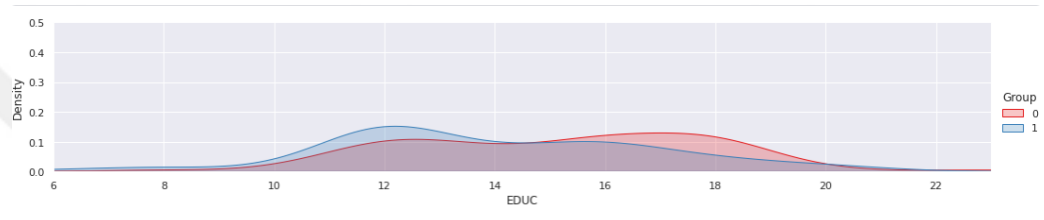


Figure 5 1 Density of Education Level Distribution

The people most likely to develop Alzheimer's disease were generally male. This suggests that men are more prone to this disease. 0 defines the patients who have no alzheimer's disease and 1 defines the patients who have alzheimer's disease.

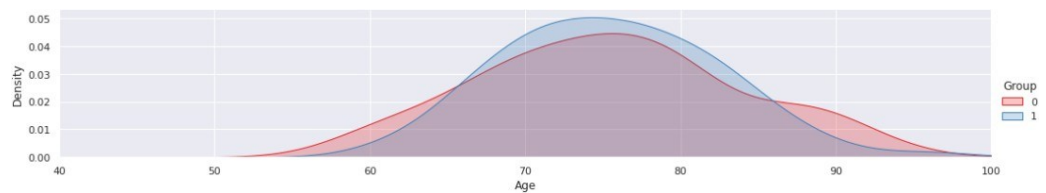


Figure 5 2 Density of Age Distribution

A mini-mental illness report, on the other hand, is an exam performed after the patient comes to the hospital for a suspected reason. As far as it is observed

intensively, the individuals who get low grades in this exam are also said to be individuals whose economic situation is bad and whose socio-economic situation is not good.

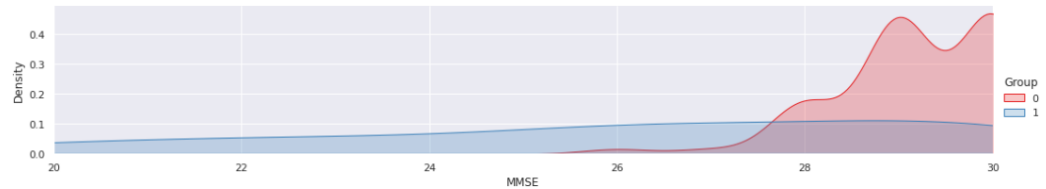


Figure 5 3 Density of Mini-Mental Exam Distribution

While the areas shown in orange in the table show the exam scores of individuals with the disease, the area missed in blue is the records of individuals without the disease.

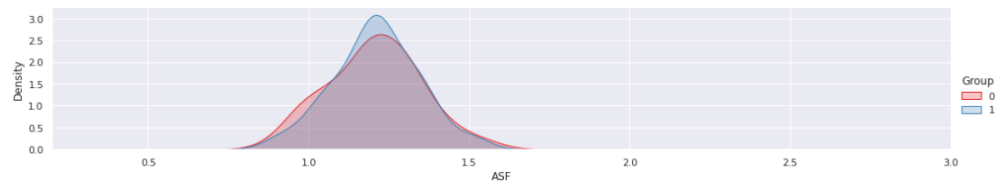


Figure 5 4 Density of Atlassian Scaling Factor Distribution

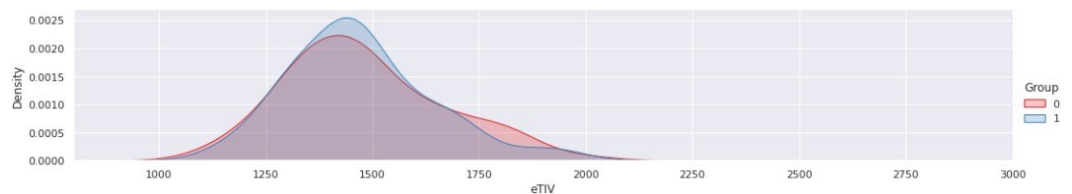


Figure 5 5 Density of Estimated Total Intracranial Volume Distribution

eITV (Estimated Total Intracranial Volume), ASF (Atlas Scaling Factor), and nWBV (Normalize Whole Brain Volume) examine brain structures and detect and monitor Alzheimer's disease from the current state of the brain. Experimental Total Intracranial Volume, abbreviated as eTIV, calculates the intracranial brain volume. Atlas Scaling Factor is the comparison of total intracranial volume with eTIV due to differences in human anatomy.

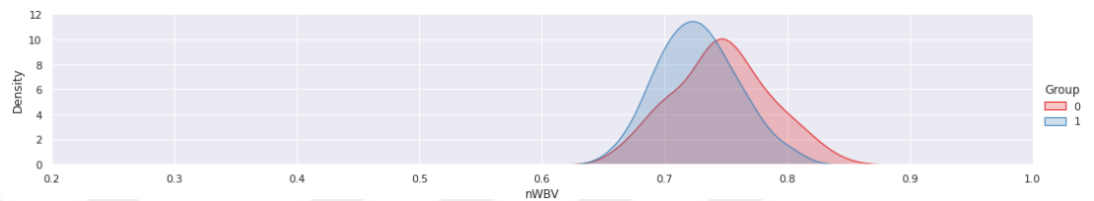


Figure 5 6 Density of Estimated Total Intracranial Volume Distribution

We can conclude that the size of the intracranial volume is quite high in ill individuals from the graphs below. This graphic briefly tells us that individuals with Alzheimer's disease have a smaller intracranial brain volume, larger cavities, and indirectly more brain folds than normal individuals. In light of known information today, it is also stated that the intracranial structures of individuals with Alzheimer's disease generally have more brain folds, a small brain volume, and a large cranial cavity.

	Model	Accuracy	Recall	AUC	False Positive Rate	True Positive Rate	TH
0	Logistic Regression With Imputation	0.763158	0.70	0.70	[0.0, 0.16666666666666666, 1.0]	[0.0, 0.7, 1.0]	[2, 1, 0]
1	Logistic Regression With Drop Blank Values	0.789474	0.70	0.70	[0.0, 0.11111111111111111, 1.0]	[0.0, 0.7, 1.0]	[2, 1, 0]
2	Support Vector Machines	0.842105	0.70	0.70	[0.0, 0.0, 1.0]	[0.0, 0.7, 1.0]	[2, 1, 0]
3	Decision Tree	0.815789	0.65	0.65	[0.0, 0.0, 1.0]	[0.0, 0.65, 1.0]	[2, 1, 0]
4	Random Forest	0.815789	0.75	0.75	[0.0, 0.11111111111111111, 1.0]	[0.0, 0.75, 1.0]	[2, 1, 0]
5	AdaBoost Algorithm	0.868421	0.65	0.65	[0.0, 0.0, 1.0]	[0.0, 0.65, 1.0]	[2, 1, 0]

Figure 5 7 A Detailed Result of Clinical Data

As mentioned in the relevant sections above, the researches and methods applied in this research progressed through two main areas. Since there are data from two different databases and these data do not have a common feature, it appears as two different studies progressing in two different branches, instead of a combined machine learning.

The first study proceeds through open-source data presented to developers from the OASIS database in tabular format, which includes information on clinical and social issues about individuals. Based on the data obtained thanks to OASIS and the examination of the studies in the literature, models were created, and studies were carried out with various classification algorithms (Logistic Regression, Black Trees, etc.). Although hyperparameter definitions were made to increase the accuracy rate in these studies, although it surpassed some studies in the literature, it remained under some studies. To summarize this situation as follows, the low number of observations on clinical data negatively affected the accuracy rate. Perhaps in the future, more data to be provided by OASIS or other data to be combined by developers can establish a guiding structure for those who refer to this research.

The second study proceeds through open source data presented to developers from the OASIS database in tabular format, which includes information on clinical and social issues about individuals. Based on the data obtained thanks to OASIS and the examination of the studies in the literature, models were created and studies were carried out with various classification algorithms (Logistic Regression, Decision Trees, etc.). Although hyperparameter definitions were made in order to increase the accuracy rate in these studies, although it surpassed some studies in the literature, it remained under some studies. To summarize this situation as follows, the low number of observations on clinical data negatively affected the accuracy rate. According to the study and the results, the best accuracy result ratio is given in Adaboosting algorithm within 86.34% and 0.65 recall value. Evaluating models with accuracy only, based on the results of clinical data, does not make the

research very accurate. Recall, AUC, FPR (False Positive Rate), and TPR (True Positive Rate) values should also be considered here. There are two algorithms with the highest accuracy. One is AdaBoost and the other is Support Vector Machines. However, investigating at the other metrics of the AdaBoost algorithm, it stands out that it has been overfitted. Because the Recall and area under curve values we observed in AdaBoost are higher than other algorithms and the FPR and TPR values are proportional, this is seen as the best method preferred for clinical data. Perhaps in the future, more data to be provided by OASIS or other data to be combined by developers can establish a guiding structure for those who refer to this research.

In the second study, the study was conducted from approximately 6000 images of individuals in which MRI images were taken and only axial, sagittal and coronal sections were included. These MRI images are available for download in various formats from the ADNI database. These formats can be in NiFTI, DCM, etc. formats. After these images are obtained, they should be converted into an image format that can be easily worked on. In the studies in the literature, it has been stated that the parts containing the brain volume in MRI images are clipped and the study continues of Nalçakan, (2019) implemented, feature analyzes are made on the images or directly work with deep learning architectures. In this study, although data augmentation was tried on images, overfitting problem was encountered. Therefore, this process was removed, images were labeled by one-hot label encoding and models were produced with deep learning architectures. With a certain number of epochs for each model, the data was trained and evaluated, and the results were examined. While CNN gives the best results in image sets with a rate of 81.34%, this rate can reach high accuracy in the future if the image set is more balanced or works with better data. The most powerful method is CNN for image datasets, although RNN and VGG19 were used, the results are not extremely good as CNN used. The model seems as if it is overlearned in these methods. In RNN & VGG19, the model is overfitted so the results are not quite accurate to this observation. CNN gives us better results for these image datasets for this research.

Unfortunately, the accuracy rate could not reach 100% due to the scarcity of the data sources provided by the organizations and the inability to be very competent for this project. However, thanks to the multiple methods applied to the research, the noise in the data was reduced, feature extractions were made, and we were able to keep the data ready for processing in a more usable and cleaner format. Thanks to the hyperparameters given after these preprocessing stages and the methods that will increase the accuracy before the model pieces of training, we have succeeded in increasing the success rates to higher levels than they should have been. Although many articles and research can be defined as competitors in the literature, the accuracy rate for tabular data is 87% in the study made due to data sources, while this rate is said to be around 82% for images.

One of the biggest advantages of using artificial intelligence techniques for Alzheimer's detection is the ability to analyze large amounts of data quickly and accurately. AI algorithms can process vast amounts of medical imaging and biomarker data, identifying patterns and anomalies that may be indicative of Alzheimer's disease. Another benefit of AI-powered Alzheimer's detection is its ability to detect the disease before symptoms become apparent. By identifying early signs of Alzheimer's disease, doctors can develop personalized treatment plans that can help slow the progression of the disease and improve outcomes for patients. Moreover, AI-powered tools can also help monitor disease progression, which can be especially important in clinical trials for new Alzheimer's treatments. By using AI to track disease progression, researchers can get a better understanding of how new treatments are affecting patients and make more informed decisions about future treatment development. Based on the findings, as a potential future direction, it may be worthwhile to further improve the accuracy rate beyond our current achievement. the study can contribute to reducing uncertainties and doubts that arise for both patients and their families when it comes to disease diagnosis, ultimately serving as a beneficial tool for healthcare professionals. With this research, besides adding a study to the literature in this field, it is also aimed to reintegrate ill individuals into the society and to improve the treatment processes under the control of doctors.

6. CONCLUSION

The potential to use machine learning interpretability techniques to provide more transparent and understandable outputs from the AI model was explored. It has led to the development of interpretability techniques that can help doctors and healthcare professionals understand how the artificial intelligence algorithm achieves good results with the prediction method, increase confidence in the model and lead to better clinical decision making. This means that methods such as feature importance analysis, sensitivity analysis, or visualization tools to explain the decision-making process of the model can be developed by future developers by building on this work, and they will provide a very different and beautiful lightwork to the literature.

In this study, studies were carried out on open datasets obtained digitally from the OASIS and ADNI communities. Clinical data and MRI images of the patients were collected. Preprocessing processes were performed on both clinical and MRI image sets, feature extractions were made, and results were obtained and compared on machine learning models. Both proposed methods have achieved quite good results from some studies in the literature. While the highest accuracy rate obtained in clinical data was 86.84% with the AdaBoosting model, the model with the highest accuracy rate obtained on image data was observed as CNN with 81.34%.

Much better results can be obtained by adding cumulatively to the research in this field. In future studies, a certain period could be determined for the early diagnosis of alzheimer's disease. By looking at the relevant patient record and patient data, information about the course of the disease such as X years and Y months can be presented to the patient's relatives or the patient himself. This will be a work that will revolutionize both the medical community and the field of artificial intelligence and data science. It is possible to continue this study in this way. These studies can be varied and evaluated in another way to help human lives and the quality of life for further research in the future.

The findings of this study suggest that the use of artificial intelligence techniques for Alzheimer's disease detection holds significant promise for improving early detection and intervention for the disease. The results of the study demonstrate that AI algorithms can accurately detect the disease at an early stage, potentially allowing for earlier treatment and better outcomes for patients. Moreover, the AI model developed in this study also has the potential to be used for disease monitoring, which could be valuable in clinical trials and other research studies. However, there are several limitations that must be acknowledged. One limitation of the study is the relatively small sample size used for model training and testing. While the results are promising, further studies with larger sample sizes are needed to confirm the findings. Additionally, the AI model developed in this study was based on a specific set of biomarkers and imaging data, which may not generalize to other populations or datasets. Further studies are needed to investigate the generalizability of the model and its performance in other settings. Despite these limitations, the results of this study provide strong evidence for the potential of AI-powered Alzheimer's disease detection and monitoring.



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İZMİR

07/04/2023

Yaren GUNDUZ



RESUME

PROFESSIONAL EXPERIENCE

- **BorgWarner Inc. – Software Tools Engineer;** Turkey (TR)
 - *June 2021 – February 2023*
- **Keonok – Data Scientist;** Poland (PL)
 - *November 2020 – May 2021*
- **Amatis.NL – Software Test Engineer & Data Scientist;** Turkey (TR)
 - *January 2021 – May 2021*

EDUCATION

- **Ege University, Master of Science (MSc)** | Turkey (TR) | June 2023
 - Concentrations: Data Science, Artificial Intelligence & Machine Learning | GPA: 3.54 / 4.0 |
 - High Honor Student
- **VBO & Miuul, Specialist Training Program** | Switzerland (CH) | June 2022
 - Concentrations: Data Science, Artificial Intelligence & Machine Learning, CRM Analytics, Recommendation Systems, RDBMS
 - Online live and mentor-supported Data Scientist Training program. 10+ real-life data science projects were implemented.
- **Yasar University, Bachelor of Science (BSc)** | Turkey (TR) | June 2020
 - Concentrations: Data Science, Artificial Intelligence and Machine Learning, Cryptography | GPA: 3.51 / 4.0
 - High Honor Student | Government Success Scholarship | Academic Success Scholarship